Positivity of $Q_{n,c}(q)$ for Cylindric Partitions: Quasi-symmetric Functions, Descent Statistics, and the Landscape of Approaches

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Abstract

We investigate Warnaar's Conjecture 2.7: the positivity of polynomials $Q_{n,c}(q)$ arising from rank-3 cylindric partition generating functions. The paper is organised in two parts. **Part I** (Cool Concepts) develops the techniques as a self-contained toolbox: quasi-symmetric functions and L-positivity, descent statistics (charge, cocharge, 0-Hecke), P-partitions, the Macdonald (q,t)-generalisation, the crush operator, and the crossing-path model on the cylinder. **Part II** (Open Problems) applies these tools to the conjecture: the ground-set formulation (S, U, ν), structural obstacles (the disconnect between L-positivity and Q-positivity, partial quasi-symmetry), partial results (cyclic descent injection for $n = 1$), the vertex-model strategy for a (q,t)-Macdonald statistic, the Rains–Warnaar (q,t)-LR conjectures, dead ends, and the most promising future directions.

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1 Definitions and setup

1.1 Cylindric plane partitions

Definition 1.1 (Gessel–Krattenthaler [5])**.** For any binary string $\pi = (\pi_1, \dots, \pi_T)$ of length T, a *cylindric plane partition* (CPP) with profile π is a sequence of integer partitions

$$\mathfrak{c} = (\mu^0, \mu^1, \dots, \mu^T), \qquad \mu^0 = \mu^T, \tag{1}$$

satisfying, for each $k = 1, \dots, T$:

- if $\pi_k = 1$, then μ^k/μ^{k-1} is a *horizontal strip*
- if $\pi_k = 0$, then μ^{k-1}/μ^k is a horizontal strip.

The condition $\mu^0 = \mu^T$ is the *cylindric closure*. When $\mu^0 = \mu^T = \varnothing$, this reduces to a *reverse plane partition* (RPP) of shape determined by π [26].

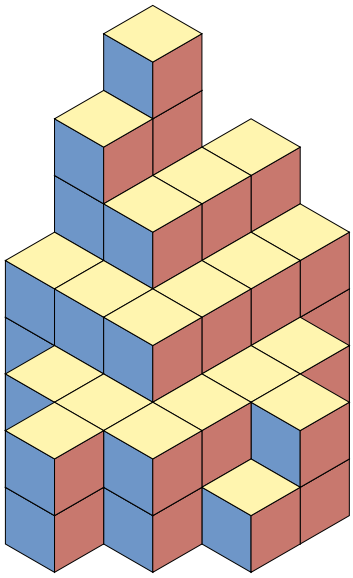
The *weight* is $|\mathfrak{c}| = |\mu^1| + |\mu^2| + \cdots + |\mu^T|$ (the boxes of μ^0 are excluded to avoid double-counting), and the *maximum entry* is $n(\mathfrak{c}) = \max\{k : \text{some cell of the labelled diagram has label } k\}$.

Remark 1.2 (Two visualisations). A CPP admits two equivalent pictorial representations.

1. **Labelled cylindric diagram.** The cells of the cylindric shape (the periodic Young diagram with profile π) carry integer labels which are weakly decreasing along both cylindric rows and cylindric columns. The partition sequence (1) is read off the NW–SE diagonals from right to left.

	4	3		
6	4	3	2	
5	4	3	2	2
3	3	3	2	1
	2	2	2	0
		2	0	0

2. **Stacked cubes (3D).** A cell with label k represents k unit cubes stacked vertically. The result is a "cylindric pile of cubes" analogous to the box-stacking picture for ordinary plane partitions [2].



Remark 1.3 (Rank-3 specialisation). The present research focuses on profiles relevant to the A_2 *Andrews–Gordon identities*. These are parametrised by a triple $c = (c_0, c_1, c_2)$ with $d = c_0 + c_1 + c_2$. The profile string has length $T = d$ and encodes the cyclic interlacing pattern among three partitions $\nu^{(1)}, \nu^{(2)}, \nu^{(3)}$.

The two-variable generating function is

$$\mathrm{GK}_c(z, q) = \sum_{\mathfrak{c} \in \mathrm{CPP}(c)} z^{n(\mathfrak{c})} q^{|\mathfrak{c}|} = \sum_{n \geq 0} g_n(q)\, z^n. \tag{2}$$

Setting $z = 1$ recovers Borodin's product formula (Section 1.3).

1.2 The NILP model

A CPP $\mathfrak{c}$ with max entry n corresponds to n *non-intersecting lattice paths* (NILPs) on the cylinder, sandwiched between two fixed boundary paths. The k-th path ($k = 1, \ldots, n$) traces the level set $\{\text{cells with label} \geq k\}$. These n paths are *free* (determined by $\mathfrak{c}$), while the boundary above (the *profile path*, determined by π) and the boundary below (the *zero-level*, where all labels vanish) are fixed. Thus there are $n + 2$ paths in total, but only n are variable. In short:

$$\textbf{max entry} = n \quad \Longleftrightarrow \quad n \text{ free non-intersecting lattice paths.}$$

1.3 Borodin's product formula and Schur positivity

Borodin [2] proved a cylindric analog of the hook-product formula:

$$\mathrm{GK}_c(1, q) = \prod_{s \in \widehat{\lambda}(\pi)} \frac{1}{1 - q^{h_{\widehat{\lambda}}(s)}}, \tag{3}$$

where $\widehat{\lambda}(\pi)$ is the cylindric diagram with profile π and $h_{\widehat{\lambda}}(s) = j - i + kT$ is the *cylindric hook length* of the cell with cylindric inversion coordinates (i, j, k) (see `tex/definitions.tex`).

This generalises Stanley's hook-product formula for reverse plane partitions [26]: $\sum_{\mathfrak{c} \in \mathrm{RPP}(\lambda)} q^{|\mathfrak{c}|} = \prod_{s \in \lambda} 1/(1 - q^{h_\lambda(s)})$. In the cylindric case, the hooks "wrap around" the cylinder, and an additional factor $\prod_{n \geq 1} 1/(1 - q^{nT})$ (counting the base partition γ) appears.

1.4 The trace formula and Schur expansion

Let $\pi = (\pi_0, \ldots, \pi_{T-1})$ be the binary profile string ($T = |\pi|$), and define the Pieri operators $G^0(u) = \Omega[Xu]$ (adds horizontal strips), $G^1(u) = \Omega^*[Xu]$ (removes horizontal strips), and the degree operator $D_w S_\lambda = w^{|\lambda|} S_\lambda$. The GK generating function is a *trace* over the Schur basis [15]:

$$\mathrm{GK}_c = \sum_\mu \langle S_\mu \,|\, G^{\pi_0}(u_0)\, G^{\pi_1}(u_1) \cdots G^{\pi_{T-1}}(u_{T-1})\, D_w \,|\, S_\mu \rangle, \tag{4}$$

where u_k, w are monomials in the refined variables $z_0, \ldots, z_{T-1}$. Inserting resolutions of the identity $\sum_\lambda |S_\lambda\rangle\langle S_\lambda|$ between each operator yields a sum over cyclic chains of partitions $\lambda^{(0)}, \ldots, \lambda^{(T-1)}$ with $\lambda^{(T)} = \lambda^{(0)}$, where each term is a product of skew Schur functions:

$$\prod_{k=0}^{T-1} \langle S_{\lambda^{(k)}} | G^{\pi_k}(u_k) | S_{\lambda^{(k+1 \bmod T)}} \rangle = \prod_k s_{\mathrm{skew}_k}(u_k).$$

By the Littlewood–Richardson rule, each such product is **Schur-positive**. The Schur coefficient $\langle s_\kappa, \mathrm{GK}_c \rangle$ is thus a sum of products of LR coefficients over all valid cyclic chains of intermediate partitions—not a clean closed form, but a definite combinatorial quantity.

Question 1.4. Give a direct combinatorial description of the Schur coefficients $\langle s_\kappa, \mathrm{GK}_c \rangle$ that does not go through iterated LR sums over cyclic chains.

Partial answer. The crossing-path model (Section 7) reformulates $\langle s_\kappa, \mathrm{GK}_c \rangle$ as a single count of crossing-path configurations on the cylinder (Proposition 7.2), eliminating the intermediate partitions. Defining statistics on these configurations is the next step.

Definition 1.5 (Cylindric skew diagram)**.** Let $\lambda, \mu \in A^+_{k,n}$ (partitions with at most k parts, each $\le n$). For $d \ge 0$, the (n,k)*-restricted cylindric skew diagram of degree* d is the infinite set of lattice cells

$$\lambda/d/\mu \;:=\; \big\{\, (i,j) \in \mathbb{Z}^2 : \lambda[d]_i \ge j > \mu[0]_i \,\big\},$$

where $\lambda[d]$ denotes the bi-infinite *cylindric loop* obtained by shifting the outline of λ by the period vector $\Omega = (k, -n)$, then translating by $(d, 0)$. Concretely, $\lambda[d]$ has a fundamental region with parts $(n^d, \lambda_1, \ldots, \lambda_k)$. The diagram is Ω-periodic: translating by $(k, -n)$ maps it to itself. For $d = 0$, this recovers the ordinary skew diagram λ/μ.

COULD YOU DRAW ME A DIAGRAM HERE. I DON"T GET IT

Definition 1.6 (Cylindric Schur function)**.** A *semistandard cylindric tableau* of shape $\lambda/d/\mu$ is a map $T : \lambda/d/\mu \to \mathbb{N}$ satisfying:

1. **Periodicity:** $T(i,j) = T(i+k,\, j-n)$ for all $(i,j) \in \lambda/d/\mu$;
2. **Weak row increase:** $T(i,j) \le T(i,j+1)$ when $(i,j+1) \in \lambda/d/\mu$;
3. **Strict column increase:** $T(i,j) < T(i+1,j)$ when $(i+1,j) \in \lambda/d/\mu$.

The weight $\mathbf{x}^T = \prod_i x_{T(i,j)}$ is taken over one fundamental period ($k + d$ consecutive rows). The *cylindric Schur function* is

$$s_{\lambda/d/\mu}(x_1, x_2, \ldots) = \sum_T \mathbf{x}^T,$$

summed over all semistandard cylindric tableaux T of shape $\lambda/d/\mu$. This is a symmetric function (McNamara [21]; see also Korff [13], Postnikov [23]).

Remark 1.7. A semistandard cylindric tableau of shape $\lambda/d/\mu$ is the same object as a CPP with the corresponding profile π. Given a CPP $(\mu^0, \ldots, \mu^T)$, label each cell of the horizontal strip $\mu^k \setminus \mu^{k-1}$ (when $\pi_k = 1$) with entry k; the result is a semistandard cylindric tableau. Conversely, reading the entry sets of a cylindric tableau by value recovers the partition sequence. The profile π determines the cylindric shape (see Definition 1.5).

Remark 1.8. This Schur positivity of GK_c should be distinguished from the question of whether individual cylindric Schur functions $s_{\lambda/d/\mu}$ are Schur-positive. McNamara [21] showed that cylindric Schur functions can have **negative** Schur coefficients. The full generating function GK_c, being a sum over all cylindric shapes, is positive by the trace/LR argument above, even though individual summands need not be.

1.5 Local rules: motivation from the Cauchy identity

The *local rule* is the combinatorial engine behind the growth-diagram proof of Borodin's identity. Its existence is motivated by the **Cauchy identity**:

$$\sum_{\lambda} S_\lambda[X_n]\, S_\lambda[Y_n] = \prod_{i,j=1}^{n} \frac{1}{1 - x_i y_j}.$$

THERE IS MORE DETAIL IN DEFINITIONS.TEX INCLUDE IT

An inductive proof (adding one variable at a time) reduces this to matching coefficients of $S_\alpha[X_{n-1}]\, S_\beta[Y_{n-1}]$ on both sides. On the RHS, after factoring out $1/(1 - x_n y_n)$, one sums over pairs (m, μ) with $\mu \in D(\alpha) \cap D(\beta)$ and $m \geq 0$; on the LHS, one sums over $\lambda \in U(\alpha) \cap U(\beta)$. The identity holds if and only if there exists a weight-preserving bijection between these two sets.

Definition 1.9 (Local rule)**.** Let $D(\alpha)$ be the set of partitions obtained from α by removing a horizontal strip, and $U(\alpha)$ the set obtained by adding one. A *local rule* is a bijection

$$\mathfrak{U}_{\alpha,\beta} : \big(\mathbb{Z}_{\geq 0} \times (D(\alpha) \cap D(\beta))\big) \xrightarrow{\sim} U(\alpha) \cap U(\beta),$$

satisfying $|\lambda| = m + |\alpha| + |\beta| - |\nu|$ whenever $\mathfrak{U}_{\alpha,\beta}(m, \nu) = \lambda$. The inverse is $\mathfrak{D}_{\alpha,\beta}(\lambda) = (m, \nu)$.

There are exactly **two** natural choices of local rule (see [15], Section 3.2):

1. **RSK rule**: corresponds to the Robinson–Schensted–Knuth correspondence;
2. **Burge rule**: corresponds to the Burge correspondence.

Both satisfy the weight condition and the commutativity property $\mathfrak{L}_i \circ \mathfrak{L}_j = \mathfrak{L}_j \circ \mathfrak{L}_i$ when $|i - j| \geq 2$, which ensures that the growth diagram is well-defined regardless of the order in which inversions are processed.

ALSO INCLUDE THE EXPLICIT RSK RULE

Explicit Burge rule

We describe the **Burge local rule** explicitly (see [15], Section 3.2).

Down operator $\mathfrak{D}_{\alpha,\beta}(\lambda) = (m, \mu)$:

1. Let $\overline{A} = \{i : \lambda'_i > \alpha'_i\}$ and $\overline{B} = \{i : \lambda'_i > \beta'_i\}$ (columns of λ strictly longer than those of α, β).
2. For each $i \in \overline{A} \cap \overline{B}$ (in decreasing order), assign $\delta(i)$ to be the largest column index $< i$ not in $\overline{A} \cup \overline{B}$ and not already assigned. Set $\overline{C} = \{\delta(i) > 0 : i \in \overline{A} \cap \overline{B}\}$.

3. Obtain μ from λ by removing a box from each column in $\overline{A} \cup \overline{B} \cup \overline{C}$. Set $m = \#\{\delta(i) \leq 0\}$.

Up operator $\mathfrak{U}_{\alpha,\beta}(m, \mu) = \lambda$:

1. Let $A = \{i : \alpha'_i > \mu'_i\}$ and $B = \{i : \beta'_i > \mu'_i\}$.
2. For each $i \in A \cap B$ (in increasing order), assign $\epsilon(i)$ to be the smallest column index $> i$ not in $A \cup B$ and not already assigned. Set $C = \{\epsilon(i)\}$.
3. Let D be the first m elements of $\mathbb{Z}_{>0} \setminus (A \cup B \cup C)$. Obtain λ from μ by adding a box to each column in $A \cup B \cup C \cup D$.

Example 1.10 (Burge rule)**.** Take $\alpha = (6, 5, 5, 3)$, $\beta = (6, 6, 5, 2)$, $\lambda = (7, 6, 5, 3, 1)$. Then:

$$\lambda' = (5, 4, 4, 3, 3, 2, 1), \quad \overline{A} = \{1, 6, 7\}, \quad \overline{B} = \{1, 3, 7\}.$$
$$\overline{A} \cap \overline{B} = \{1, 7\}.$$

Processing $i = 7$: $\delta(7) = 5$ (largest < 7 not in $\{1, 3, 6, 7\}$). Processing $i = 1$: $\delta(1) = 0$ (no column < 1 is available). So $\overline{C} = \{5\}$, $m = 1$, and μ is obtained by removing a box from columns $\{1, 3, 5, 6, 7\}$, giving $\mu = (6, 5, 4, 2)$.

$$\boxed{\mathfrak{D}_{(6,5,5,3),(6,6,5,2)}(7, 6, 5, 3, 1) = \big(1,\ (6, 5, 4, 2)\big).}$$

The inverse $\mathfrak{U}_{\alpha,\beta}(1, (6, 5, 4, 2)) = (7, 6, 5, 3, 1)$ follows by the dual greedy matching.

1.6 The cylindric growth diagram

A *cylindric growth diagram* with profile π is a labelling of the cylindric poset (the quotient $\mathbb{Z}^2/(x, y) \sim (x + n, y - m)$) by integer partitions at vertices and nonneg integers at faces, such that:

1. adjacent partitions along edges differ by a horizontal strip;
2. all sufficiently deep vertices carry the same label γ;
3. the local rule $\mathfrak{U}_{\alpha,\beta}(m, \nu) = \lambda$ holds at every face.

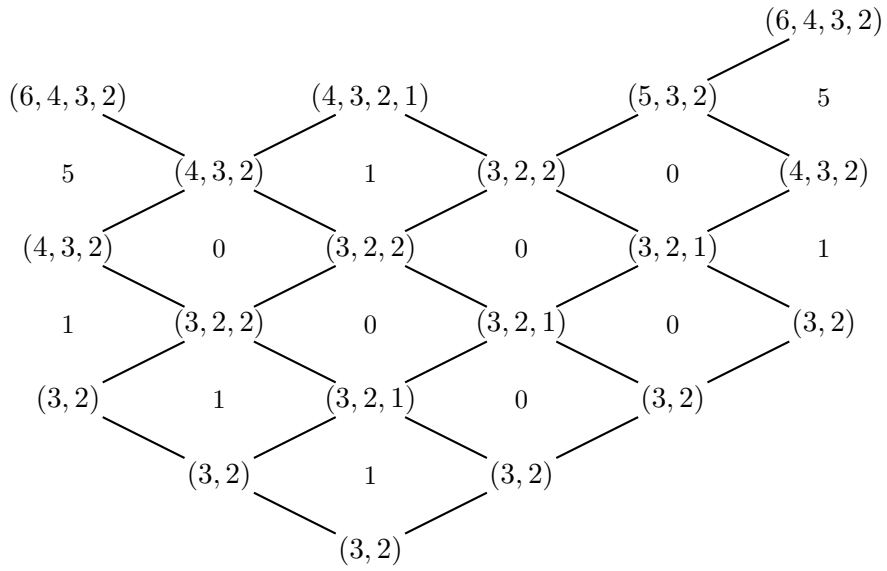
The upper boundary reads off a CPP $\mathfrak{c} \in \mathrm{CPP}(\pi)$; the face labels assemble into an arbitrarily labelled cylindric diagram $\mathfrak{d} \in \mathrm{ALCD}(\pi)$. By the invertibility of the local rule, this gives a *strongly weight-preserving* bijection [15]:

$$\psi_\pi : (\mathrm{IP}, \mathrm{ALCD}(\pi)) \xrightarrow{\sim} \mathrm{CPP}(\pi), \tag{5}$$

satisfying $|\mu^k| = |\gamma| + |\mathrm{diag}(k)|_{\mathfrak{d}}$ for each step k.

This proves Borodin's identity: CPPs biject with (partition γ, labelled cylindric diagram $\mathfrak{d}$), and the hook-product formula counts the latter. The growth diagram mirrors the algebraic proof step-by-step: each row corresponds to one Pieri operator in the trace formula (4), and the vertex labels are the intermediate partitions.

Example 1.11 (Growth diagram)**.** Below is a growth diagram for profile $\pi = 00101$ (i.e. $c = (2, 1, 2)$, $d = 5$). Vertex labels are partitions; face labels are nonneg integers consumed by the local rule. The bottom row repeats the base partition $\gamma = (3, 2)$ (periodically on the cylinder); the top vertex gives the CPP entries.



The face labels on each row encode the CPP entries at that level. The left and right ends of each row are identified on the cylinder.

Remark 1.12 (Two bijections and Schützenberger). The RSK and Burge choices yield bijections ψ_R, ψ_B. Their composition $\psi_R \circ \psi_B^{-1} : \mathrm{CPP}(\pi) \to \mathrm{CPP}(\pi)$ is a *Schützenberger-type involution* on CPPs, preserving the profile and the weight. Its representation-theoretic meaning at the level of Tingley's $\widehat{\mathfrak{sl}}_n$ crystal [28] is an open question.

Part I
Cool Concepts

2 Quasi-symmetric functions and L-positivity

2.1 The frequency generating function

For a CPP π, define the value frequency $\#_k(\pi) = |\{(i,j) : \nu_j^{(i)} = k\}|$. The *frequency generating function* is

$$\widetilde{\mathrm{GK}}_c(\mathbf{z}) = \sum_{\pi \in \mathrm{CPP}(c)} z_1^{\#_1(\pi)} z_2^{\#_2(\pi)} \cdots \in \mathbb{Z}[[z_1, z_2, \ldots]]. \tag{6}$$

Under $z_k \mapsto q^k$, this recovers $\mathrm{GK}_c(1, q)$.

Observation 2.1 (Verified: 55 profiles, $d \le 5$)**.** $\widetilde{\mathrm{GK}}_c$ is quasi-symmetric: $\widetilde{\mathrm{GK}}_c = \sum_\alpha a_\alpha(c)\, M_\alpha$ with $a_\alpha \in \mathbb{Z}_{\ge 0}$. It is *not* symmetric ($a_{(2,1)} \ne a_{(1,2)}$ in general).

The quasi-symmetry follows from the observation that the interlacing conditions (1) depend only on the *relative order* of entries, not their absolute values. This is an instance of Stanley's P-partition theory:

By the Stanley–Gessel theory of P-partitions (Section 4), the generating function of any labeled poset is quasi-symmetric and L-positive (Theorem 4.6). The CPP interlacing conditions are precisely of P-partition type (weak and strict comparisons on entries), and hence $\widetilde{\mathrm{GK}}_c$ is quasi-symmetric.

2.2 L-positivity

Remark 2.2 (L-positivity as open problem). While L-positivity of $\widetilde{\mathrm{GK}}_c$ is computationally verified, a **proof** remains open. The P-partition theorem gives L-positivity for any *fixed* max entry (the poset is acyclic after unrolling the cycle), but $\widetilde{\mathrm{GK}}_c$ sums over *all* max entries, and it is not obvious that an infinite sum of L-positive functions remains L-positive.

Remark 2.3 (Symmetrisation). Under the natural projection $\mathrm{QSym} \to \mathrm{Sym}$ given by $M_\alpha \mapsto m_{\mathrm{sort}(\alpha)}$ (equivalently $L_\alpha \mapsto s_{\mathrm{sort}(\alpha)}$), L-positivity implies Schur positivity. Concretely, $\widetilde{\mathrm{GK}}_c$ maps to $\sum_\alpha b_\alpha(c)\, s_{\mathrm{sort}(\alpha)}$, grouping compositions by their underlying partition. The symmetrisation of $\widetilde{\mathrm{GK}}_c$ is thus a Schur-positive symmetric function, recovering the Schur positivity of GK_c from a finer quasi-symmetric structure.

The *fundamental quasi-symmetric functions* L_α (also denoted F_α) are a second basis of QSym:

$$L_\alpha = \sum_{\beta \preceq \alpha} M_\beta,$$

where $\beta \preceq \alpha$ means β refines α. The M-to-L conversion uses Möbius inversion: $b_\alpha = \sum_{\beta \succeq \alpha} (-1)^{\ell(\beta)-\ell(\alpha)} a_\beta$.

Observation 2.4 (Verified: 55 profiles, $d \le 5$)**.** $\widetilde{\mathrm{GK}}_c$ is L-positive: $\widetilde{\mathrm{GK}}_c = \sum_\alpha b_\alpha(c)\, L_\alpha$ with $b_\alpha \in \mathbb{Z}_{\ge 0}$.

L-positivity is strictly stronger than M-positivity (which is automatic since a_α counts CPPs). Stabilised L-coefficients exhibit product structure: $c = (1,1,0)$ gives $b_\alpha^{\mathrm{stab}} = 2^{\ell(\alpha)}$; $c = (2,1,1)$ gives $5^{\ell(\alpha)}$.

3 Descent statistics: charge, cocharge, and 0-Hecke

3.1 Reading words and the Littlewood–Richardson rule

Both the Littlewood–Richardson rule and the charge statistic depend on a *reading word*: a linear ordering of the cells of a (skew) tableau.

Definition 3.1 (Reading word)**.** The *reverse reading word* of a (skew) tableau T is the sequence of entries obtained by reading each row from **right to left**, starting from the **top row** and proceeding downward. (Equivalently: columns from right to left, top to bottom within each column.)

Definition 3.2 (Lattice word)**.** A word $w = w_1 w_2 \cdots w_N$ in the alphabet $\{1, 2, \ldots, k\}$ is a *lattice word* (or *ballot sequence*, or *Yamanouchi word*) if for every prefix $w_1 \cdots w_j$ and every letter i,

$$\#\{m \le j : w_m = i\} \;\ge\; \#\{m \le j : w_m = i{+}1\}.$$

In words: at every point in the word, we have seen at least as many i's as $(i{+}1)$'s, for all i.

Theorem 3.3 (Littlewood–Richardson rule)**.** *$s_\mu \cdot s_\nu = \sum_\lambda c_{\mu\nu}^\lambda\, s_\lambda$, where the* Littlewood–Richardson coefficient *$c_{\mu\nu}^\lambda$ counts the number of semistandard skew tableaux of shape λ/μ and content ν whose reverse reading word is a lattice word.*

Example 3.4 (LR coefficients for $s^2_{(2,1)}$)**.** Compute $c^{(3,2,1)}_{(2,1),(2,1)}$: fill the skew shape $(3,2,1)/(2,1)$ with content $(2,1) = \{1,1,2\}$. The skew cells are $(1,3)$, $(2,2)$, $(3,1)$ (one per row, all in different columns, so no row/column constraints between them). All three assignments are valid SSYT:

Filling	$(1,3),(2,2),(3,1)$	Rev. reading word	Lattice?
T_1	1, 1, 2	1, 1, 2	**Yes**
T_2	1, 2, 1	1, 2, 1	**Yes**
T_3	2, 1, 1	2, 1, 1	No: prefix "2" has $\#_2 > \#_1$

T_3 fails at the very first letter: the prefix "2" has one 2 but zero 1's. Thus $c^{(3,2,1)}_{(2,1),(2,1)} = 2$.

Key point. The reading word turns a 2D object (a skew tableau) into a 1D word. The lattice condition selects which fillings "count." The same idea—choose a reading order, then check a combinatorial property of the resulting word—is what we seek for CPPs.

3.2 Charge, cocharge, and Kostka–Foulkes polynomials

Definition 3.5 (Kostka–Foulkes polynomial)**.** Let $\lambda \vdash n$ and $\mu \vdash n$. The *Kostka–Foulkes polynomial* $K_{\lambda,\mu}(q)$ is the transition coefficient between the Schur basis and the Hall–Littlewood basis: $s_\lambda = \sum_\mu K_{\lambda,\mu}(q)\, P_\mu(x;q)$. Equivalently, $K_{\lambda,\mu}(q)$ is the coefficient of s_λ in $\sum_T q^{\mathrm{charge}(T)}$, summed over all $T \in \mathrm{SSYT}(\lambda,\mu)$.

Definition 3.6 (Charge statistic)**.** The *charge* of Lascoux–Schützenberger [17] is a statistic charge : $\mathrm{SSYT}(\lambda,\mu) \to \mathbb{Z}_{\geq 0}$ defined as follows. Given a semistandard tableau T of shape λ and content μ:

1. Form the *reading word* of T: read each row from **left to right**, starting from the **bottom row** and proceeding upward. (For a standard tableau this gives a permutation; for a semistandard tableau, a word with repeated letters.)

2. Decompose the reading word into *standard subwords* (each containing exactly one copy of each letter $1,\ldots,k$).

3. For each standard subword σ (a permutation of $\{1,\ldots,k\}$), define $\mathrm{index}(1) = 0$ and, for $i = 2,\ldots,k$:

$$\mathrm{index}(i) = \begin{cases} \mathrm{index}(i{-}1) & \text{if letter } i \text{ appears to the } \textit{right} \text{ of letter } i{-}1 \text{ in } \sigma, \\ \mathrm{index}(i{-}1) + 1 & \text{otherwise.} \end{cases}$$

   Set $\mathrm{charge}(\sigma) = \sum_{i=1}^{k} \mathrm{index}(i)$.

4. The charge of T is the sum of charges of its standard subwords.

The *cocharge* is $\mathrm{cocharge}(T) = n(\lambda,\mu) - \mathrm{charge}(T)$, where $n(\lambda,\mu) = \sum_{i<j} \min(\mu_i, \lambda'_j)$.

Example 3.7 (Computing charge)**.** Take $\lambda = (2,1)$, $\mu = (1,1,1)$. Since $\mu = 1^3$, each SSYT is a standard Young tableau. There are two:

$$T_1 = \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 \\ \cline{1-1} \end{array} \qquad T_2 = \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1} \end{array}$$

Reading words (bottom to top, left to right): $w(T_1) = 3\,1\,2$ and $w(T_2) = 2\,1\,3$.

Charge of T_1 $(\sigma = 3\,1\,2)$:

- index(1) = 0.
- Letter 2 is at position 3, letter 1 at position 2: 2 is to the *right* of 1 ⇒ index(2) = 0.
- Letter 3 is at position 1, letter 2 at position 3: 3 is to the *left* of 2 ⇒ index(3) = 0 + 1 = 1.

charge(T_1) = 0 + 0 + 1 = 1.

Charge of T_2 ($\sigma = 2\,1\,3$):

- index(1) = 0.
- Letter 2 at position 1, letter 1 at position 2: 2 is to the *left* of 1 ⇒ index(2) = 1.
- Letter 3 at position 3, letter 2 at position 1: 3 is to the *right* of 2 ⇒ index(3) = 1.

charge(T_2) = 0 + 1 + 1 = 2.

Therefore $K_{(2,1),(1^3)}(q) = q^1 + q^2 = q + q^2$.

The key theorem is:

$$K_{\lambda,\mu}(q) = \sum_{T \in \mathrm{SSYT}(\lambda,\mu)} q^{\mathrm{charge}(T)} \in \mathbb{N}[q].$$

The nonnegativity is highly nontrivial (proved via the theory of crystal bases, or equivalently via Haiman's proof using the $n!$-conjecture).

We verified computationally that $Q_{n,c}(q)$ does *not* decompose as a nonneg sum of Kostka–Foulkes polynomials. However, the L-positivity of $\widetilde{\mathrm{GK}}_c$ suggests a *different* charge-like statistic may exist on CPPs, one adapted to compositions rather than partitions. See `scripts/analyze_Q.py`.

Question 3.8. Does there exist a charge-like statistic on CPPs (or on (S, U, ν) triples) that explains the positivity of $Q_{n,c}(q)$?

Question 3.9. Is there a (q,t)-analog of the conjecture? The Hall–Littlewood case ($t = 0$ in the Macdonald framework) should recover the original conjecture. Replacing the Schur basis by the Macdonald basis $\{P_\lambda(x; q, t)\}$ in the trace formula gives a (q,t)-weighted GK function (Section 5); does the Q-polynomial extraction yield a polynomial with nonneg coefficients in q and t?

Partial answer (Section 5.3): The P/Q-basis formulation gives *rational functions* of q, t, not polynomials, so the question is ill-posed in that basis. The modified Macdonald basis $\tilde{H}_\mu$ does give polynomial Schur coefficients (Haiman), but reformulating the Borodin identity in $\tilde{H}$ is open.

3.3 0-Hecke modules

The 0-*Hecke algebra* $H_n(0)$ is the quotient of the free algebra on generators $T_1, \ldots, T_{n-1}$ by the Coxeter braid relations and the idempotent relations $T_i^2 = T_i$. Its irreducible modules are one-dimensional and indexed by compositions of n.

By the work of Tewari–van Willigenburg [27], L-positivity of a quasi-symmetric function f is equivalent to f being the *quasi-symmetric characteristic* of an $H_n(0)$-module. Thus:

Corollary 3.10. *Observation 2.4 implies the existence of a graded $H_n(0)$-module M_c such that* $\mathrm{ch}(M_c) = \widetilde{\mathrm{GK}}_c$.

Conjecture 3.11. *There exists a natural descent-type statistic $\alpha(\pi)$ on CPPs such that* $\widetilde{\mathrm{GK}}_c = \sum_\pi L_{\alpha(\pi)}$.

Finding such a statistic would give a combinatorial proof of L-positivity and could illuminate the original conjecture.

4 P-partitions

4.1 What is a P-partition?

A P-partition is a generalisation of a semistandard Young tableau. The idea: instead of filling the cells of a Young diagram, you fill the elements of a *poset* with positive integers, subject to ordering rules.

Definition 4.1 (Labeled poset)**.** A *labeled poset* is a pair (P, ω) where:

- P is a finite partially ordered set (poset), and
- $\omega : P \to \{1, 2, \ldots, |P|\}$ is a bijection (a labeling of the elements by distinct integers).

A *descent* of (P, ω) is a covering relation $x \lessdot_P y$ where $\omega(x) > \omega(y)$ (the label goes *down* while the poset goes *up*). An *ascent* is one where $\omega(x) < \omega(y)$.

Definition 4.2 ((P, ω)-partition)**.** Given a labeled poset (P, ω), a (P, ω)-*partition* is a map $\sigma : P \to \mathbb{Z}_{>0}$ such that for every covering relation $x \lessdot_P y$:

- **At an ascent** ($\omega(x) < \omega(y)$): require $\sigma(x) \leq \sigma(y)$ (weakly increasing),
- **At a descent** ($\omega(x) > \omega(y)$): require $\sigma(x) < \sigma(y)$ (strictly increasing).

In plain language: the poset P tells you which elements must be compared. The labeling ω tells you whether each comparison is weak ($\leq$) or strict ($<$).

Example 4.3 (SSYT as a P-partition)**.** A semistandard Young tableau of shape $\lambda = (2, 2)$ is a filling of four cells with positive integers, rows weakly increasing left to right, columns strictly increasing bottom to top (French convention). This is a P-partition where:

The poset P has four elements arranged in a 2×2 grid, with covering relations $(1, 1) \lessdot (1, 2)$ (same row), $(2, 1) \lessdot (2, 2)$ (same row), $(1, 1) \lessdot (2, 1)$ (same column), $(1, 2) \lessdot (2, 2)$ (same column). The labeling ω is chosen so that:

- Row neighbours are ascents (ω increases left to right) $\Rightarrow$ weak inequality $\leq$,
- Column neighbours are descents (ω decreases bottom to top) $\Rightarrow$ strict inequality $<$.

For instance: $\omega(1, 1) = 2$, $\omega(1, 2) = 4$, $\omega(2, 1) = 1$, $\omega(2, 2) = 3$. Then $(1, 1) \lessdot (1, 2)$: $\omega = 2 < 4$ (ascent $\Rightarrow \leq$, rows weakly increasing). And $(1, 1) \lessdot (2, 1)$: $\omega = 2 > 1$ (descent $\Rightarrow <$, columns strictly increasing).

Example 4.4 (A non-tableau P-partition)**.** Consider a poset P on three elements $\{a, b, c\}$ with a single relation $a <_P b$ (so c is incomparable to both). Label: $\omega(a) = 2$, $\omega(b) = 1$, $\omega(c) = 3$.

Since $\omega(a) = 2 > 1 = \omega(b)$ and $a \lessdot_P b$, this is a *descent*, so any (P, ω)-partition must have $\sigma(a) < \sigma(b)$ (strict). The value $\sigma(c)$ is unconstrained (since c is incomparable to everything).

Some valid (P, ω)-partitions: $(\sigma(a), \sigma(b), \sigma(c)) = (1, 2, 1),\ (1, 3, 5),\ (2, 4, 1)$.
But $(3, 3, 1)$ is **not** valid because $\sigma(a) = 3 \not< 3 = \sigma(b)$.

4.2 The generating function and L-positivity

Given a labeled poset (P, ω), define its *P-partition generating function*:

$$\Gamma_{(P,\omega)}(\mathbf{z}) = \sum_{\sigma} \prod_{p \in P} z_{\sigma(p)},$$

where the sum is over all (P, ω)-partitions $\sigma : P \to \mathbb{Z}_{>0}$.

Example 4.5 (Generating function of Example 4.4)**.** For $P = \{a, b, c\}$ with $a <_P b$ (descent) and c incomparable:

$$\Gamma_{(P,\omega)} = \sum_{\substack{i<j \\ k \geq 1}} z_i \, z_j \, z_k = \Big(\sum_{i<j} z_i z_j\Big)\Big(\sum_{k \geq 1} z_k\Big).$$

This is quasi-symmetric: if we set $z_1 = z_2 = \cdots = z_n = 1$ and all others to 0, the result depends only on n (it equals $\binom{n}{2} \cdot n$), not on *which* n variables we chose.

The fundamental theorem is:

Theorem 4.6 (Stanley 1972, Gessel 1984)**.** *For any labeled poset (P, ω):*

1. *$\Gamma_{(P,\omega)}$ is quasi-symmetric.*
2. *$\Gamma_{(P,\omega)}$ is L-positive: $\Gamma_{(P,\omega)} = \sum_{\alpha} b_\alpha L_\alpha$ with $b_\alpha \in \mathbb{Z}_{\geq 0}$.*
3. *The coefficient b_α counts the number of* linear extensions *of P whose* descent composition *equals α.*

The key notion is:

Definition 4.7 (Linear extension and descent composition)**.** A *linear extension* of a poset P on elements $\{p_1, \ldots, p_n\}$ is a listing $\pi = \pi_1, \pi_2, \ldots, \pi_n$ of the elements of P such that if $x <_P y$ then x appears before y in π. (It is a "reading word": a way to read all elements of P in an order compatible with the partial order.)

The *descent set* of a linear extension π (with respect to the labeling ω) is

$$\mathrm{Des}(\pi) = \{i : \omega(\pi_i) > \omega(\pi_{i+1})\} \subseteq \{1, 2, \ldots, n-1\}.$$

The *descent composition* is the composition of n obtained by recording the lengths of the runs between descents. For instance, if $n = 5$ and $\mathrm{Des} = \{2, 4\}$, the descent composition is $(2, 2, 1)$.

Example 4.8 (Linear extensions and L-coefficients)**.** Continuing Example 4.4: $P = \{a, b, c\}$ with $a <_P b$ and c incomparable, $\omega(a) = 2$, $\omega(b) = 1$, $\omega(c) = 3$.

The linear extensions of P must list a before b; c can go anywhere:

Extension π	Labels $(\omega(\pi_1), \omega(\pi_2), \omega(\pi_3))$	Des	Composition
a, b, c	2, 1, 3	$\{1\}$	$(1, 2)$
a, c, b	2, 3, 1	$\{2\}$	$(2, 1)$
c, a, b	3, 2, 1	$\{1, 2\}$	$(1, 1, 1)$

So $\Gamma_{(P,\omega)} = L_{(1,2)} + L_{(2,1)} + L_{(1,1,1)}$.

Each L-coefficient counts linear extensions with a given descent composition. **This is what "L-positivity for free" means**: the L-coefficients are always nonneg because they literally count something.

5 The Macdonald (q,t)-generalisation

5.1 The (q,t)-Borodin identity

The trace formula (4) generalises to Macdonald polynomials [15]: replace the Hall inner product $\langle -,-\rangle$ by the Macdonald inner product $\langle -,-\rangle_{q,t}$, the Schur basis $\{S_\lambda\}$ by $\{P_\lambda, Q_\lambda\}$, and the Pieri operators Ω, Ω^* by $\Omega_{q,t}, \Omega^*_{q,t}$:

$$\sum_{\mathfrak{c}\in \mathrm{CPP}(\pi)} W_{\mathfrak{c}}(q,t)\, z^{|\mathfrak{c}|} = \sum_{\mu} \langle Q_\mu \,|\, H^{\pi_0}(u_0)\cdots H^{\pi_{T-1}}(u_{T-1})\, D_w \,|\, P_\mu\rangle_{q,t}, \tag{7}$$

where $H^0(u) = \Omega_{q,t}[Xu]$, $H^1(u) = \Omega^*_{q,t}[Xu]$, and the Macdonald weight is

$$W_{\mathfrak{c}}(q,t) = \prod_{k:\pi_k=1} \varphi_{\mu^k/\mu^{k-1}}(q,t) \prod_{k:\pi_k=0} \psi_{\mu^{k-1}/\mu^k}(q,t).$$

At $q=t$ the Pieri coefficients equal 1, $P_\lambda = Q_\lambda = S_\lambda$, and one recovers the Schur case.

The proof telescopes identically: the Macdonald commutation relation gives the factor $(tu_iu_jw^{m+1};q)_\infty/(u_iu_jw^{m+1};q)_\infty$ at each step (replacing $1/(1-u_iu_jw^{m+1})$ from the Schur case), and the boundary term is the same $\prod_{n\geq 1} 1/(1-w^n)$.

5.2 Combinatorial weight via peaks and valleys

The weight function has a combinatorial description in the NILP model [15]:

$$W_{\mathfrak{c}}(q,t) = \Omega\big[(q-t)\,\mathcal{D}_{\mathfrak{c}}(q,t)\big], \qquad \mathcal{D}_{\mathfrak{c}} = \sum_{s\in \mathrm{peak}} q^{a(s)}t^{\ell(s)} - \sum_{s\in \mathrm{valley}} q^{a(s)}t^{\ell(s)}, \tag{8}$$

where peak/valley cubes are defined by the local shape of the associated lattice path (up-down vs. down-up), and $a(s), \ell(s)$ are the arm and leg lengths of the cube s in its host partition.

The Hall–Littlewood case ($q=0$): only arm-length-zero cubes contribute, and the weight reduces to a product over surface cubes weighted by t^{level}, recovering the formula of Corteel–Savelief–Vuletić [4].

5.3 Schur coefficients are rational functions (computed)

Observation 5.1 (Verified computationally, $|\mu| \leq 4$)**.** The Schur expansion of Macdonald P_μ and Q_μ has coefficients that are **rational functions** of q,t, not polynomials:

1. $P_\mu \to s$: the transition coefficients are $K_{\lambda,\mu}(q,t)/b_\mu(q,t)$, which are *rational* in q,t. Example: $\langle s_{(1,1)}, P_{(2)}\rangle = (-q+t)/(qt-1)$.
2. $Q_\mu \to s$: also rational (same phenomenon).
3. Products $P_\lambda \cdot Q_\mu \to s$: rational.
4. $\sum_{|\mu|=n} Q_\mu \to s$: rational.
5. Skew $Q_{\lambda/\mu} \to s$: rational.
6. Products of skew Q functions $\to s$: rational.

The denominators are products of cyclotomic-type factors $(1-q^at^\ell)$ arising from the normalisation $b_\mu(q,t) = \prod_{s\in\mu} \frac{1-q^{a_\mu(s)+1}t^{\ell_\mu(s)}}{1-q^{a_\mu(s)}t^{\ell_\mu(s)+1}}$ that distinguishes P_μ from $\tilde{H}_\mu$.

5.4 Why the coefficients are rational: the commutation alphabet

The rationality has a clean structural explanation through the plethystic calculus. There are two layers:

Layer 1: Pieri alphabet (polynomial). The weight function (8) is $W_{\mathfrak{c}} = \Omega[(q-t)\mathcal{D}_{\mathfrak{c}}]$, where $\mathcal{D}_{\mathfrak{c}} = \sum_s \pm q^{a(s)}t^{\ell(s)}$ is a *polynomial* in q, t (a signed sum of monomials over peaks/valleys). But Ω converts each letter $(q-t)\cdot q^a t^\ell = q^{a+1}t^\ell - q^a t^{\ell+1}$ into the *ratio*

$$\Omega[q^{a+1}t^\ell - q^a t^{\ell+1}] = \frac{1-q^a t^{\ell+1}}{1-q^{a+1}t^\ell},$$

which is the Pieri coefficient formula [18].

Layer 2: Commutation alphabet (rational). The Macdonald commutation relation $\Omega^*_{q,t}[Xu]\,\Omega_{q,t}[Xv] = \frac{(tuv;q)_\infty}{(uv;q)_\infty}\,\Omega_{q,t}[Xv]\,\Omega^*_{q,t}[Xu]$ introduces a scalar factor at each step of the telescoping proof. Plethystically, this factor is

$$\frac{(tuv;q)_\infty}{(uv;q)_\infty} = \Omega\left[-\frac{1-t}{1-q}\cdot uv\right]. \tag{9}$$

The "internal alphabet" $\frac{1-t}{1-q}$ is a **rational function** of q, t. This is the fundamental source of the rationality: each factor in the Borodin product carries this rational kernel, which propagates to all z-coefficients and thence to the Schur expansion.

At $q = t$: $\frac{1-t}{1-q} = 1$, so $\frac{(quv;q)_\infty}{(uv;q)_\infty}$ telescopes to $\frac{1}{1-uv}$—a geometric series with *integer* coefficients. This is why the Schur case gives nonneg integers while the Macdonald case gives rational functions.

Clearing denominators. The modified Macdonald polynomial $\tilde{H}_\mu = t^{n(\mu)}c_\mu\, P_\mu$ multiplies by $c_\mu(q,t) = \prod_{s\in\mu}(1-q^{a+1}t^\ell)/(1-q^a t^{\ell+1})$, which is precisely the accumulated product of the Ω-produced denominators. After clearing, the Schur coefficients become polynomials (Haiman).

5.5 Modified Macdonald polynomials: Haiman positivity

In contrast, the *modified Macdonald polynomials* $\tilde{H}_\mu$ have polynomial nonneg Schur coefficients (Haiman's theorem [7]):

$$\tilde{H}_\mu = \sum_\lambda \tilde{K}_{\lambda,\mu}(q,t)\, s_\lambda, \qquad \tilde{K}_{\lambda,\mu}(q,t) \in \mathbb{N}[q,t]. \tag{10}$$

This was confirmed computationally for $|\mu| \le 4$. (See `scripts/macdonald_schur.py`.)

Remark 5.2 (Consequence for the Borodin identity). The (q,t)-Borodin identity (7) uses the P/Q basis (through Pieri operators and the Macdonald inner product). Since the $P/Q \to s$ transition is rational, the Macdonald-weighted GK generating function does **not** have polynomial Schur coefficients in the standard P/Q formulation. The original question ("is the Macdonald GK Schur-positive?") is thus **ill-posed** in the P/Q basis: one gets rational functions, and asking for polynomial nonnegativity does not make sense.

Question 5.3. Can the (q,t)-Borodin identity be reformulated in the $\tilde{H}_\mu$ basis, where Schur coefficients *are* polynomials? Specifically, define

$$\mathrm{GK}_c^{\tilde{H}}(z;q,t) = \sum_\mu \langle \tilde{H}_\mu \,|\, (\text{Pieri chain}) \,|\, \tilde{H}_\mu \rangle_{q,t}\, z^{|\mu|}.$$

Do the Schur coefficients of $\mathrm{GK}_c^{\tilde{H}}$ lie in $\mathbb{N}[q,t]$? This would require understanding the $\tilde{H}$-Pieri rule (which involves creation operators and is more complex than the P/Q Pieri rule).

6 The crush operator

6.1 Crush on NILPs

Given a CPP with max entry n (equivalently, n non-intersecting lattice paths on the cylinder), the *crush* of level k ($1 \le k \le n$) replaces every entry equal to k with $k-1$. In the NILP model, this pushes the k-th path flush against the $(k-1)$-th path above it.

The result is a CPP with the same profile but with max entry $\le n$ (the k-th and $(k-1)$-th paths may merge). Crucially, **crush does not preserve weight**: the weight decreases by $\#_k(\pi)$, the number of cells carrying value k, which varies from CPP to CPP.

6.2 Crush as an algebraic operator on QSym

On the frequency generating function, crush acts cleanly:

$$\mathrm{crush}_k : z_k \mapsto z_{k-1} \quad \text{(shift value index down by 1)}.$$

In infinitely many variables, this is an automorphism of QSym. However, in the finite alphabet $z_1, \ldots, z_n$ it *escapes*: crush_1 sends $z_1 \mapsto z_0$, which does not exist.

The crush operator is *indexed by* $n-j$ (removing the j-th level set from the bottom), but the number of boxes removed depends on the CPP. This makes it fundamentally difficult to build weight-preserving bijections from crush: **it is hard to find a combinatorial operator that adds or removes exactly k boxes** uniformly across all CPPs of a given profile. Crystal operators (Kashiwara $\tilde{e}_i, \tilde{f}_i$; see Tingley [28]) change weight by a simple root α_i (one box), but they do not preserve the max entry. The tension between controlling the max entry and controlling the weight is the core obstacle.

7 The crossing-path model on the cylinder

The trace formula (4) computes $\langle s_\kappa, \mathrm{GK}_c \rangle$ by inserting resolutions of the identity between each Pieri operator, producing a sum over cyclic chains of intermediate partitions $\lambda^{(0)} \to \cdots \to \lambda^{(T-1)} \to \lambda^{(0)}$, where each step contributes an LR coefficient. The result is a *sum of products* of LR coefficients—manifestly nonneg but opaque.

The key observation is that the intermediate partitions are **artifacts of slicing the cylinder into T strips**. Each LR coefficient $c^{\nu}_{\lambda\mu}$ counts configurations of *intersecting lattice paths* in a triangular (or rectangular) region: two families of paths cross each other, and the boundary data (λ, μ, ν) are read from the edges of the region. When two adjacent strips share a boundary partition $\lambda^{(k)}$, the outgoing paths from one strip become the incoming paths for the next.

Gluing. Rather than slicing the cylinder into T strips and summing over intermediate partitions, we *glue* the strips into a single cylindric region and let both families of paths run all the way around:

- **Family 1 (horizontal paths).** These are the partition-boundary paths. At any column of the growth diagram, the partition $\lambda^{(k)}$ determines a lattice path (the boundary of its Young diagram). As k varies around the cylinder, this path *evolves* through the T profile steps. In the glued picture, these paths wrap horizontally around the cylinder.

- **Family 2 (vertical paths).** At each profile step k, the Pieri operator adds or removes a horizontal strip. The cells in this strip determine a second family of paths, running vertically

through the growth diagram. These encode the *shape data* — which cells are added/removed at each step.

The two families **cross** at each cell of the growth diagram. At each crossing, the Burge local rule (Section 1.5) resolves the interaction: it determines how the horizontal path deforms in response to the vertical path passing through.

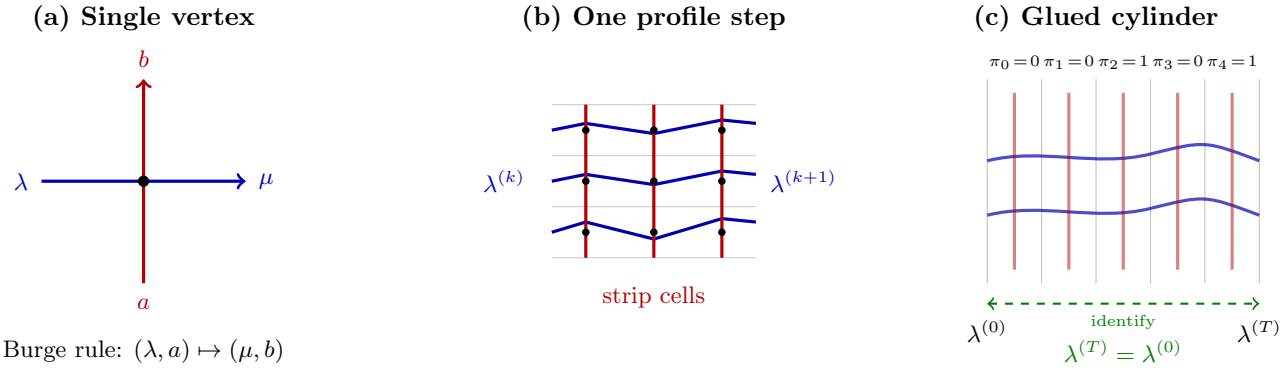


Figure 1: The crossing-path model. **(a)** A single vertex: a horizontal path (blue, partition boundary) crosses a vertical path (red, strip cell). The Burge local rule determines the outgoing states (μ, b) from the incoming states (λ, a). **(b)** One profile step: multiple horizontal paths (evolving partition boundaries) cross multiple vertical paths (cells of the horizontal strip). **(c)** The glued cylinder ($T = 5$ steps, profile $\pi = 00101$): both families run all the way around; the left and right boundaries are identified. The intermediate partitions $\lambda^{(1)}, \ldots, \lambda^{(T-1)}$ are not part of the input—they emerge from the path configuration.

Definition 7.1 (Crossing-path configuration)**.** A *crossing-path configuration* on the cylinder with profile π consists of:

1. a family of horizontal paths wrapping around the cylinder (encoding the evolving partition boundaries), and
2. a family of vertical paths threading through the growth diagram (encoding the strip data at each profile step),

subject to the Burge crossing rule at every intersection point, and the **periodic boundary condition**: the horizontal paths close up after T steps (i.e. $\lambda^{(T)} = \lambda^{(0)}$).

Proposition 7.2. *The Schur coefficient of* GK_c *is a* single count*:*

$$\langle s_\kappa, \mathrm{GK}_c \rangle = \#\{\textit{crossing-path configurations on the cylinder with output shape } \kappa\},$$

where the "output shape" is read from the vertical paths (the total shape of the cells added across all profile steps).

This eliminates the intermediate partitions from the description: they are determined by the paths at the strip boundaries, and summing over all cyclic chains is equivalent to summing over all valid path configurations in the interior.

Note that individual cylindric Schur functions $s_{\lambda/d/\mu}$ are **not** always Schur-positive (McNamara [21]), but the full $\mathrm{GK}_c = \sum s_{\lambda/d/\mu}$ is, because the crossing-path count is manifestly nonneg.

Remark 7.3 (Vertex model interpretation). The crossing-path model is a *vertex model* on the cylinder. Each crossing of a horizontal path with a vertical path is a *vertex* with a local state determined by the incoming path segments. The Burge rule assigns admissibility (Boltzmann weight 1 or 0) to each vertex. The periodic boundary condition makes GK_c a *transfer matrix trace*, and the Schur coefficient $\langle s_\kappa, \mathrm{GK}_c \rangle$ is the partition function restricted to configurations with winding data κ.

This connects to Korff's integrable model for cylindric partitions [13], where GK is realised as a partition function of the $\widehat{\mathfrak{sl}}_r$ vertex model, and to the R-matrix framework of Section 12.

7.1 Local rules, R-matrices, and Littlewood–Richardson coefficients

The crossing-path model (Definition 7.1) is our growth-diagram language for the same object that Zinn-Justin [34] studies from the integrable-systems side. This subsection makes the dictionary explicit: the local rule *is* an R-matrix, commutativity of local rules *is* the Yang–Baxter equation, and the LR coefficient $c_{\mu\nu}^{\lambda}$ *is* the partition function of a vertex model on a triangle.

The tiling model on the triangular lattice

Zinn-Justin works on a triangular lattice with two species of colored paths:

- **Green paths** (our "horizontal" family): move one half-step to the right at each time step. In our language, these are the evolving partition boundaries.
- **Red paths** (our "vertical" family): move one half-step to the left at each time step. In our language, these encode the strip data (which cells are added/removed at each profile step).

Each edge of the lattice carries a state from $\{-, +, 0\}$ (empty, green, red), and each vertex is a *tile* whose admissibility and weight are determined by the four surrounding edge states.

The key tiles:

Tile type	Description	Weight
$\alpha_\pm$	Same-color lines propagate straight through	1
$\beta_\pm$	Opposite-color lines cross	1
$\gamma_\pm$	A line moves in the "wrong" direction (interaction)	u
γ_0	Green and red pass through each other transparently	1

In the unweighted (Schur) case, $u = 0$: the interaction tiles $\gamma_\pm$ are forbidden, and the model reduces to free propagation with crossings—precisely the admissibility condition of our Burge local rule.

The R-matrix

The Boltzmann weights at each vertex assemble into an **R-matrix** $R(x)$: a linear map $R(x) : V \otimes V \to V \otimes V$ where $V = \mathbb{C}^3$ (the three edge states $\{-, +, 0\}$). The matrix is mostly $\{0, 1\}$-valued; the only entry depending on the spectral parameter x is

$$R_{(-,+)}^{(-,+)}(x) = x,$$

corresponding to the interaction tile γ. Setting $x = 0$ recovers the **bijective** local rule: each legal input (α, β) has exactly one legal output (λ, m) with nonzero weight.

The Yang–Baxter equation is commutativity of local rules

The R-matrix satisfies the **Yang–Baxter equation** (star-triangle relation): for spectral parameters satisfying $x + y + z = 0$,

$$R_{12}(x)\, R_{13}(y)\, R_{23}(z) = R_{23}(z)\, R_{13}(y)\, R_{12}(x).$$

This is the integrable-systems version of the commutativity property $\mathfrak{L}_i \circ \mathfrak{L}_j = \mathfrak{L}_j \circ \mathfrak{L}_i$ for $|i-j| \geq 2$ (Section 1.5).

In the growth-diagram proof of Borodin's identity (Section 1.6), this commutativity ensures that the growth diagram is *well-defined* regardless of the order in which rows are processed. In Zinn-Justin's language, the same property ensures that the *transfer matrices commute*:

$$[\widetilde{T}_+(u),\, \widetilde{T}_-(v)] = 0, \qquad [\widetilde{T}_\pm(u),\, T] = 0, \qquad [\widetilde{T}_+(u),\, \widetilde{T}_+(v)] = 0.$$

The proof is the standard "RTT = TTR" argument: insert an R-matrix at one end of a product of two transfer matrices, use the Yang–Baxter equation to push it across row by row, and remove it at the other end.

Our growth diagram	**Zinn-Justin's vertex model**
Vertex of the grid	Vertex of the triangular lattice
Input partitions (α, β)	Input edge states
Output (λ, m)	Output edge states
Burge rule $\mathfrak{U}_{\alpha,\beta}(m, \mu) = \lambda$	R-matrix entry $R^{(\lambda,m)}_{(\alpha,\beta,\mu)}$
Weight $\lvert\lambda\rvert = m + \lvert\alpha\rvert + \lvert\beta\rvert - \lvert\mu\rvert$	Boltzmann weight
Commutativity ($\lvert i-j\rvert \geq 2$)	Yang–Baxter equation
Growth diagram = bijection CPP $\leftrightarrow$ ALCD	Transfer matrix trace = partition function
Schur, all weights $\in \{0, 1\}$	$x = 0$: free fermion limit

LR coefficients as a partition function on a triangle

The payoff of Zinn-Justin's framework is a direct, algebraic proof that $c^\lambda_{\mu\nu}$ counts puzzle tilings. The argument (Section 5 of [34]) proceeds in five steps:

1. **Start from the supersymmetric Schur function.** Express $s_\lambda(u_1, \ldots, u_m / v_1, \ldots, v_n)$ as a transfer-matrix expectation value:

$$s_\lambda(\mathbf{u}/\mathbf{v}) = \langle \varnothing \,|\, \textstyle\prod_i \widetilde{T}_+(u_i)\, \prod_j \widetilde{T}_-(v_j) \,|\, \lambda \rangle.$$

2. **Evolve the state $|\lambda\rangle$ forward.** Apply the (parameter-free) transfer matrix T repeatedly. After $2k$ steps (for k large enough), the state decomposes into spatially separated components:

$$T^{2k}|\lambda\rangle = \sum_{\mu,\nu} c^\lambda_{\mu\nu}\, |\mu\rangle \sqcup_k |\nu\rangle,$$

   where $|\mu\rangle \sqcup_k |\nu\rangle$ means the green particles encoding μ have moved far to the left and the red particles encoding ν have moved far to the right. The coefficients $c^\lambda_{\mu\nu}$ are defined by this expansion.

3. **Use commutativity to separate.** Since all transfer matrices commute (Yang–Baxter!), move T^{2k} past the $\widetilde{T}_{\pm}$ operators:

$$s_\lambda(\mathbf{u}/\mathbf{v}) = \sum_{\mu,\nu} c^\lambda_{\mu\nu} \, \langle \varnothing \mid \textstyle\prod_i \widetilde{T}_+(u_i) \, \prod_j \widetilde{T}_-(v_j) \mid \mu \rangle \sqcup_k |\nu\rangle.$$

4. **Factor into independent Schur functions.** Since $|\mu\rangle$ and $|\nu\rangle$ are spatially separated, the expectation value factors:

$$s_\lambda(\mathbf{u}/\mathbf{v}) = \sum_{\mu,\nu} c^\lambda_{\mu\nu} \, s_\nu(u_1, \ldots, u_m) \, s_{\mu^{\mathrm{T}}}(v_1, \ldots, v_n).$$

5. **Identify the coefficients.** The LHS is the supersymmetric Schur function, whose expansion in products of ordinary Schur functions is unique. Therefore $c^\lambda_{\mu\nu}$ *are* the Littlewood–Richardson coefficients.

Remark 7.4 (The triangle). The evolution $T^{2k}|\lambda\rangle$ takes place entirely within a finite triangular region of the lattice: green particles drift left, red particles drift right, and after enough steps they separate. The Young diagrams μ and ν are read from the binary sequences of particles on the two upper edges of the triangle. Therefore $c^\lambda_{\mu\nu}$ counts the number of valid **tilings of the triangle** with boundary data (λ, μ, ν)—these are precisely the *puzzles* of Knutson–Tao [9].

Remark 7.5 (Why this matters for us). The crossing-path model on the cylinder (Proposition 7.2) writes $\langle s_\kappa, \mathrm{GK}_c \rangle$ as a count of path configurations. Each "strip" of the cylinder contributes an LR coefficient, which by the above is a puzzle count. The full cylindric generating function is thus a **transfer matrix trace** over puzzle partition functions.

This is the starting point for the (q,t)-deformation programme (Section 12): replace the $\{0,1\}$-valued Boltzmann weights with polynomial (q,t)-weights, turning counting into a weighted generating function. The Yang–Baxter equation guarantees that the resulting (q,t)-LR coefficients are well-defined (independent of the lattice decomposition), and if the Boltzmann weights are nonneg polynomials, positivity follows for free.

Part II
Open Problems

8 Warnaar's Conjecture and the ground-set formulation

8.1 The Q-polynomial

Definition 8.1. The *Q-polynomial* is

$$Q_{n,c}(q) = (q;q)_n \cdot [z^n]\big((zq;q)_\infty \cdot \mathrm{GK}_c(z,q)\big). \tag{11}$$

Conjecture 8.2 (Warnaar [29], Conjecture 2.7)**.** *For all $n \geq 0$ and all rank-3 profiles c with $d \not\equiv 0 \pmod 3$, one has $Q_{n,c}(q) \in \mathbb{N}[q]$.*

Remark 8.3 (Why rank 3?). Conjecture 8.2 is the A_2 case ($r = 3$) of a general conjecture for all ranks. The Andrews–Gordon identities for A_1 ($r = 2$) are classical, and the Q-polynomials reduce to q^{n^2} or $q^{n(n+1)}$ (see Section 15). The A_2 case is the first genuinely open case.

Conjecture 8.4 (Warnaar [29], Conjecture 8.4: general rank)**.** *Fix $d, r \geq 1$ and a profile $c = (c_0, \ldots, c_{r-1})$ of level d (i.e. $c_0 + \cdots + c_{r-1} = d$). Then $Q_{n,c}(q) \in \mathbb{N}[q]$ and*

$$Q_{n,c}(1) = \left(\frac{\gcd(d,r)}{d+r} \binom{d+r}{r} - \gcd(d,r) \right)^n . \tag{12}$$

Remark 8.5 (Recovering the A_2 case). For $r = 3$ and $\gcd(d,3) = 1$, formula (12) gives $Q_{n,c}(1) = \left(\frac{1}{d+3}\binom{d+3}{3} - 1\right)^n = \left(\frac{(d+1)(d+2)}{6} - 1\right)^n$, matching Conjecture 8.2. For $r = 4$, $d = 7$: $Q_{n,c}(1) = (\frac{1}{11}\binom{11}{4} - 1)^n = 29^n$. The polynomiality and the $q = 1$ evaluation have been established by Welsh [30]; the **nonnegativity of coefficients** remains open.

Remark 8.6 (The $d \equiv 0 \pmod 3$ exclusion). Warnaar excludes $d \equiv 0 \pmod 3$ from the conjecture. Computationally, the cyclic descent injection (Section 14) fails for $d \equiv 0 \pmod 3$ because the balanced slack $(d/3, d/3, d/3)$ has empty cyclic descent set. For $d = 3$, the triple count per weight oscillates $(3, 3, 4, 3, 3, 4, \ldots)$, preventing any uniform injection. However, direct computation (see below) shows that $Q_{n,c}(q)$ is *still nonneg* for $d \equiv 0 \pmod 3$; the issue is that $Q_n(1)$ is no longer a fixed power K^n independent of the profile.

Observation 8.7. For all six profiles tested with $d = 3$ and $d = 6$ ($n \leq 3$, $\max_w = 20$), $Q_{n,c}(q)$ has nonneg integer coefficients. Representative data ($d = 3$, $c = (1,1,1)$):

n	$Q_{n,c}(q)$	$Q_n(1)$
0	1	1
1	$2q + 2q^2 + 3q^3$	7
2	$q^3 + 3q^4 + 4q^5 + 7q^6 + \cdots + 3q^{12}$	49
3	$2q^7 + 2q^8 + 6q^9 + \cdots$	277

The crucial difference from $d \not\equiv 0 \pmod 3$: $Q_3(1) = 277 \neq 7^3 = 343$, and $Q_3(1)$ depends on the profile ($(2,1,0) \to 218$, $(3,0,0) \to 134$).

Computationally verified for $d = 1, 2, 4, 5, 7$ (10 profiles, all pass). For $d = 2$: Q_n is a monomial—q^{n^2} (first Rogers–Ramanujan) or $q^{n(n+1)}$ (second RR identity).

8.2 Derivation: the (S, U, ν) ground set

ALSO DO THE DERIVSTION OF THE (S, ν) GROUND SET EXPANDING IN TERMS OF THE h_m.

WHAT IS THE QUASI-SYMMETRIC ANALOG OF THE h_m

Expanding the three factors in (11)—Euler's identity for $(zq;q)_\infty$, the q-binomial interpretation of $(q;q)_n/(q;q)_j$, and the subset expansion of $(q;q)_m$—yields:

Proposition 8.8.

$$Q_{n,c}(q) = \sum_{j=0}^{n} \sum_{\substack{S \subseteq \{1,\ldots,n\} \\ |S|=j}} \sum_{U \subseteq \{1,\ldots,n-j\}} \sum_{\substack{\nu \in \mathrm{CPP}(c) \\ \max(\nu) = n-j}} (-1)^{|S|+|U|}\, q^{\sigma(S)+\sigma(U)+|\nu|}, \tag{13}$$

where $\sigma(S) = \sum_{s \in S} s$.

Proof sketch. Use $q^{j(j+1)/2}\begin{bmatrix} n \\ j \end{bmatrix} = \sum_{|S|=j} q^{\sigma(S)}$ and $(q;q)_m = \sum_{U \subseteq [m]} (-1)^{|U|} q^{\sigma(U)}$, then combine with $g_m = \sum_{\max(\nu)=m} q^{|\nu|}$. □

The sign is $(-1)^{|S|+|U|}$; the conjecture asserts that the total signed weight at each q-power is nonneg. The three components play distinct roles:

- $S \subseteq \{1, \ldots, n\}$ with $|S| = j$: selects which "paths" (level sets) to remove; the q-binomial $q^{j(j+1)/2}\begin{bmatrix} n \\ j \end{bmatrix}$ counts these subsets by $\sigma(S)$.
- $U \subseteq \{1, \ldots, n-j\}$: the $(q)_{n-j}$ factor creates an alternating sum over subsets of remaining paths.
- $\nu \in \mathrm{CPP}(c)$ with $\max(\nu) = n - j$: the cylindric partition with $n - j$ paths.

8.3 The q-binomial decomposition

Define $h_m = (q;q)_m \cdot g_m$ (the "corrected" coefficients). Then:

$$Q_n = \sum_{j=0}^{n} (-1)^j \, q^{\binom{j+1}{2}} \begin{bmatrix} n \\ j \end{bmatrix} h_{n-j}, \tag{14}$$

with inverse

$$h_m = \sum_{k=0}^{m} q^k \begin{bmatrix} m \\ k \end{bmatrix} Q_{m-k}. \tag{15}$$

The h_m are nonneg polynomials for all tested profiles ($d = 1, 2, 4, 5, 7$). By (15), $Q_n \geq 0$ implies $h_m \geq 0$ (all terms are nonneg); the converse fails because (14) is an alternating sum.

Remark 8.9 (h_m positivity is combinatorially clear). The positivity of h_m should **not** be regarded as deep. From the NILP / growth-diagram perspective, h_m counts a subset of *arbitrarily labelled* cylindric diagrams: a CPP with max entry m is a cylindric diagram (the support) together with a labelling of its cells by values in $\{1, \ldots, m\}$ subject to interlacing conditions. Multiplying g_m by $(q;q)_m$ restricts to a finite sub-collection of such labellings (analogous to the passage from semistandard to standard fillings in the P-partition framework, where $(q)_p \cdot \Omega_P(q) = W_P(q)$ counts strict P-partitions). The result is manifestly a generating function with nonneg coefficients.

This perspective comes from the growth-diagram / Burge local-rule toolkit and may not be apparent from the q-series viewpoint, where multiplying by the alternating polynomial $(q;q)_m$ has no obvious reason to preserve positivity.

The hard problem is the alternating q-binomial sum (14): even with $h_m \geq 0$ as input, the signs $(-1)^j$ produce massive cancellation that must resolve to a nonneg polynomial Q_n. Proving Conjecture 8.2 requires explaining *this* cancellation, not the positivity of h_m.

Definition 8.10 (Drop and crush)**.** Let $\mathfrak{c}$ be a CPP with max entry n (equivalently, n non-intersecting lattice paths).

- **Drop level** k ($1 \leq k \leq n$): remove the k-th path (counting from the bottom). In the labelled diagram, replace every label $r \geq k$ with $r - 1$. The result has max entry $\leq n - 1$. On the frequency GF, drop acts as $z_r \mapsto z_{r-1}$ for all $r \geq k$.
- **Crush level** k ($1 \leq k \leq n$): push the k-th path flush against the $(k+1)$-th path above it. In the labelled diagram, replace every label equal to k with $k - 1$. The result has max entry $\leq n$ but the k-th and $(k-1)$-th paths may merge. Crush decreases the weight by $\#_k(\mathfrak{c})$ (the number of cells with value k), which **varies** across CPPs.

Remark 8.11 (NILP interpretation of (S, U)). Starting with a CPP with max entry n, the set S (with $|S| = j$) selects j paths to *drop*. This leaves a CPP with max entry $n-j$. The set $U \subseteq \{1, \ldots, n-j\}$ then selects paths to *crush*. The result is a CPP ν with $\max(\nu) = n - j$. However, crush does **not preserve weight** (each crush removes a variable number of boxes), so this map is not invertible and does not yield a sign-reversing involution.

I WILL DRAW THE DIAGRAMS FOR THIS, IT REALLY BELONGS IN THE DEAD END SECTION

8.4 Multivariate q-analog: non-positivity

A natural refinement replaces the univariate weight $q^{|\nu|}$ of a CPP with per-value variables: if ν has f_k cells of value k, set $q_\nu = q_1^{f_1} q_2^{f_2} \cdots$. The specialisation $q_k \mapsto q^k$ recovers $q^{|\nu|}$. Defining $Q_{n,c}(q_1, q_2, \ldots)$ via the corresponding multivariate substitution in (11), the coefficients of the monomials q_μ are **not nonneg** for $n \geq 2$. The positivity of $Q_{n,c}(q)$ does *not* lift to this multivariate refinement. (See `scripts/qsym_Q_polynomial.py`.)

Remark 8.12. This per-value refinement is *not* the same as Borodin's multivariable extension of the product formula, which tracks diagonal weights (one variable per diagonal of the cylindric diagram) rather than per-value frequencies.

8.5 Sharpened formulation: the (S, ν') involution problem

Since $h_m \geq 0$ is combinatorially clear (Remark 8.9), the three-level (S, U, ν) ground set (13) can be collapsed to a **two-level** problem. The U-factor is absorbed:

$$\sum_{U \subseteq [n-j]} (-1)^{|U|}\, q^{\sigma(U)} \cdot \sum_{\max(\nu) = n-j} q^{|\nu|} \;=\; (q;q)_{n-j} \cdot g_{n-j} \;=\; h_{n-j} \;\geq\; 0.$$

Write $h_m = \sum_{\nu'} q^{|\nu'|}$, where ν' ranges over the *restricted labelled cylindric diagrams* counted by h_m. Then equation (14) becomes

$$Q_n = \sum_{j=0}^{n} \sum_{\substack{S \subseteq [n] \\ |S| = j}} \sum_{\nu'} (-1)^j\, q^{\sigma(S) + |\nu'|}, \tag{16}$$

where ν' ranges over restricted labellings with max entry $n - j$. The conjecture is equivalent to the existence of a **sign-reversing involution** $(S, \nu') \mapsto (S', \nu'')$ with:

1. $|S'| = |S| \pm 1$ (to flip $(-1)^j$),
2. $\sigma(S') + |\nu''| = \sigma(S) + |\nu'|$ (weight-preserving),
3. fixed points all have even $|S|$ (so the residual sum is nonneg).

Remark 8.13 (Why this is cleaner). The three-level approach failed because the U-toggle involution produced cross-cancellation between different QSym monomials (§8.4). By absorbing the U-level into the combinatorially-transparent h_m, we avoid that issue entirely. The involution now operates only on S (which paths to drop), adjusting ν' (the restricted labelling) to compensate the weight change. In the NILP picture: adding or removing one path from S changes $\sigma(S)$ by the index of the path and changes ν' from a labelling with max entry $n - j$ to one with max entry $n - j \mp 1$.

9 The disconnect — why L-positivity does not imply Q-positivity

Despite these strong properties, the QSym structure does *not* help with Conjecture 8.2:

1. The max-entry statistic $n(\pi) = \max\{k : \#_k > 0\}$ is not a monomial specialisation of the z_k;
2. The E–H factorisation does not lift to QSym: both $\prod(1-tz_k)\cdot\widetilde{\mathrm{GK}}_c$ (with max-entry variable) and $\prod(1-z_k)\cdot\widetilde{\mathrm{GK}}_c$ (without) have negative coefficients;
3. Cross-cancellation between different frequency vectors (§8.4) is essential.

The positivity of $Q_{n,c}(q)$ is an *intrinsically one-variable* phenomenon.

10 Partial quasi-symmetry and the max-entry refinement

10.1 CPPs with max $= n$ as $\mathbb{Z}[z_1,\ldots,z_{n-1}][z_n]$

Write $\tilde{g}_n = \sum_{\max(\pi)=n} z_1^{\#_1}\cdots z_n^{\#_n}$ for CPPs with exact max entry n. This is *not* quasi-symmetric in $z_1,\ldots,z_n$ (the value n is privileged). However:

Observation 10.1 (Verified: 55 profiles, $n \le 3$)**.** Writing $\tilde{g}_n = \sum_{m\ge 1} z_n^m \cdot f_{n,m}(z_1,\ldots,z_{n-1})$, each slice $f_{n,m}$ is quasi-symmetric in $z_1,\ldots,z_{n-1}$:

$$\tilde{g}_n \in z_n \cdot \mathrm{QSym}(z_1,\ldots,z_{n-1})[z_n].$$

Slice M-positivity holds (coefficients count CPPs), but **slice L-positivity fails** at $n \ge 3$: the M-coefficients satisfy $a_{(k)} > a_{(j,k)}$, forcing $L_{(a,b)} = M_{(a,b)} - M_{(a+b)} < 0$.

10.2 Variable-count mismatch

The E–H factorisation couples objects in incompatible numbers of variables: $(q;q)_n$ lifts to $\prod_{k\ge1}(1-z_k)$ (infinitely many variables), while $\tilde{g}_n$ is a polynomial in only $z_1,\ldots,z_n$. This structural mismatch obstructs any multivariate lift of $Q_{n,c}$.

10.3 RPPs with max n and RSK

For reverse plane partitions (RPPs) of shape λ:

$$\sum_{\substack{\pi\in\mathrm{RPP}(\lambda)\\ \max(\pi)\le n}} q^{|\pi|} = s_\lambda(1,q,\ldots,q^{n-1}). \tag{17}$$

Proof sketch. An RPP of shape λ with max entry $\le n$ is the same as a semistandard Young tableau (SSYT) of shape λ with entries in $\{0,1,\ldots,n-1\}$ (with weakly increasing rows and columns, read as an RPP). By the combinatorial definition of the Schur function, $s_\lambda(x_1,\ldots,x_n) = \sum_{T\in\mathrm{SSYT}(\lambda,n)} x^T$. Setting $x_k = q^{k-1}$ for $k = 1,\ldots,n$ gives weight $q^{|\pi|}$ to each RPP π, yielding (17). □

The exact max-n count is thus $g_{n,\lambda} = s_\lambda(1,q,\ldots,q^{n-1}) - s_\lambda(1,q,\ldots,q^{n-2})$, manifestly nonneg since it counts objects (RPPs with max entry *exactly* n). Under RSK, max entry $= n$ means n rows in the growth diagram, $\max(P) = n$ in the insertion tableau, and an alphabet of size n for the Schur function.

For CPPs the analogous extraction is *not* a simple difference but involves the infinite product $(zq;q)_\infty$—this is precisely where positivity becomes hard.

11 Connecting P-partitions to CPPs

The CPP interlacing conditions (1) are precisely of P-partition type (weak and strict order comparisons on entries), *except* for the cyclic closure $\mu^T = \mu^0$.

For any **fixed max entry** $\leq n$, the poset is acyclic: the cycle can be "unrolled" since all values lie in $\{0, 1, \ldots, n\}$, and classical P-partition theory applies. The L-coefficients then count linear extensions of the poset by descent set.

This explains the quasi-symmetry and L-positivity of $\widetilde{\mathrm{GK}}_c$ *degree by degree* (i.e. for each fixed n). The full $\widetilde{\mathrm{GK}}_c$ is an *infinite sum* of these contributions, and the fact that an infinite sum of L-positive functions remains L-positive is nontrivial (it holds here because the posets are nested: the max-n poset is a subposet of the max-$(n+1)$ poset).

However, the cyclic structure is essential for the cylindric identities, and "cutting the cycle" destroys the algebraic connection to Rogers–Ramanujan-type identities.

Remark 11.1 (Why the reading word matters for CPPs). Theorem 4.6 says that L-positivity comes from counting linear extensions by descent set. A *linear extension* is a *reading word*: a way to linearly order the elements of the poset. For ordinary tableaux, the standard reading word (right to left, top to bottom) is the natural choice. For CPPs, the analogous question is: **what is the right way to linearly order the cells of a cylindric plane partition?** The cyclic structure (the profile wraps around $\mathbb{Z}/3\mathbb{Z}$) means there is no single canonical starting point, and different choices may give different descent sets. This is the content of Conjecture 3.11.

12 A (q,t)-Macdonald statistic on LR configurations

Central ambition. The Schur coefficients $\langle s_\kappa, \mathrm{GK}_c \rangle$ are products of LR coefficients (Section 1.4). A long-standing goal is to define a *Macdonald statistic*—a (q,t)-weight on the combinatorial objects that compute LR coefficients (intersecting lattice paths, puzzles, hive configurations)—such that:

- the statistic arises naturally from the *plethystic alphabet* $\Omega[(q-t)\,\mathcal{D}]$ that governs the Macdonald Pieri coefficients (Section 5);
- the weighted count of LR configurations gives a (q,t)-analog of the LR coefficient;
- the cylindric Schur coefficients—being products of these (q,t)-LR coefficients—inherit positivity.

Why this should work: the pieces already exist.

1. **The plethystic weight is local.** Each horizontal strip λ/μ in the interlacing sequence carries the Pieri weight $\varphi_{\lambda/\mu}(q,t) = \Omega[(q-t)(\mathcal{A}_{\lambda/\mu} - \mathcal{B}_{\lambda/\mu})]$ (thesis [15], Chapter 4). In the NILP model, this factorises over *individual cubes*: each cube s contributes $\frac{1-q^{a(s)}t^{\ell(s)+1}}{1-q^{a(s)+1}t^{\ell(s)}}$ depending on whether it is a peak or valley (eq. (8)). This is a **local statistic on the lattice path configuration**—exactly the kind of data an R-matrix encodes.
2. **Zinn-Justin's integrable tiling framework.** Zinn-Justin [34] showed that LR coefficients $c^{\nu}_{\lambda\mu}$ equal the partition function of an integrable tiling model: each puzzle piece (= vertex of a lattice model) carries a Boltzmann weight, and the Yang–Baxter equation ensures the count is well-defined. In the *unweighted* (Schur) case, all Boltzmann weights are 1 or 0, and the partition function simply *counts* puzzles.

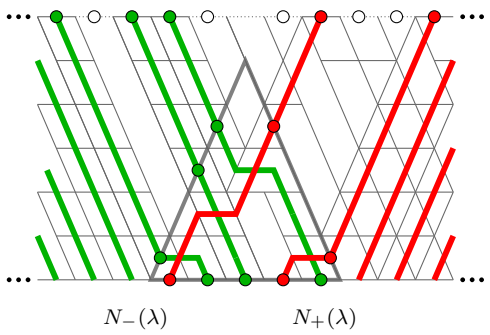
3. **The local rule *is* the R-matrix.** The Burge local rule (Definition 1.9, Section 1.5) and the R-matrix are **the same mathematical object** viewed from two sides:

Growth diagram	**Lattice model**
Vertex of the grid	Vertex of the lattice
Input partitions (α, β)	Input edge states
Output partition λ, integer m	Output edge states
Burge local rule $\mathfrak{U}_{\alpha,\beta}(m,\mu) = \lambda$	R-matrix entry $R^{(\lambda,m)}_{(\alpha,\beta,\mu)}$
Weight preservation $\lvert\lambda\rvert = m + \lvert\alpha\rvert + \lvert\beta\rvert - \lvert\mu\rvert$	Boltzmann weight
Commutativity $\mathfrak{L}_i \circ \mathfrak{L}_j = \mathfrak{L}_j \circ \mathfrak{L}_i$ for $\lvert i-j\rvert \geq 2$	Yang–Baxter equation

In the crossing-path model (Section 7), this identification is literal: the horizontal path (partition boundary) and vertical path (strip cell) cross at a vertex, and the Burge rule determines the outgoing states—exactly as an R-matrix determines scattering amplitudes.

In the Schur (unweighted) case, the local rule is a *bijection*: each legal input has exactly one legal output, so all Boltzmann weights are 0 or 1. The key insight for the (q,t)-deformation is that replacing the bijective local rule with a *weighted* R-matrix—where each vertex carries a polynomial Boltzmann weight in (q,t)—turns the growth diagram from a counting device into a *generating function machine*.

This is already done at the algebraic level. The (q,t)-Borodin identity (Section 5, eq. (7)) gives a Macdonald analog of Borodin's product formula, proved in [15]. The Macdonald Pieri coefficients $\varphi_{\lambda/\mu}(q,t)$ and $\psi_{\lambda/\mu}(q,t)$ play exactly the role of Boltzmann weights at each vertex of the growth diagram: the proof telescopes identically to the Schur case, with $1/(1-u_iu_jw^{m+1})$ replaced by $(tu_iu_jw^{m+1};q)_\infty/(u_iu_jw^{m+1};q)_\infty$. What remains open is making this identification *combinatorially explicit*: expressing $\varphi_{\lambda/\mu}$ as a product of R-matrix entries over vertices of the crossing-path model (Question 12.3), which would upgrade the algebraic identity to a vertex-model partition function and open the door to positivity via Wheeler's framework [32, 33, 1].



4. **The R-matrix gives the (q,t)-deformation.** Wheeler–Zinn-Justin [31] express Hall–Littlewood polynomials as partition functions of the t-boson model, whose R-matrix is the

trigonometric six-vertex R-matrix:

$$R_{ab}(z) = \begin{pmatrix} \frac{1-tz}{1-z} & 0 & 0 & 0 \\ 0 & t & \frac{(1-t)z}{1-z} & 0 \\ 0 & \frac{1-t}{1-z} & 1 & 0 \\ 0 & 0 & 0 & \frac{1-tz}{1-z} \end{pmatrix}, \qquad z = x/y.$$

This R-matrix weight $\frac{1-tz}{1-z}$ is precisely the commutation factor $\Omega\big[-\frac{1-t}{1-q}\cdot uv\big]$ from eq. (9). Wheeler–Zinn-Justin [32] further extend this to Hall polynomials and inverse Kostka polynomials via "right-triangle puzzles," where the puzzle pieces carry *polynomial* t-weights.

5. **The Pieri coefficient = accumulated R-matrix.** The key identification: the Macdonald Pieri coefficient $\varphi_{\lambda/\mu}(q,t)$ should equal the *accumulated product of R-matrix Boltzmann weights* along the horizontal strip λ/μ. That is, the plethystic algebra $\Omega[(q-t)(\mathcal{A}-\mathcal{B})]$ from the thesis and the integrable R-matrix from the six-vertex model are **the same object**, viewed respectively from the algebraic (Macdonald inner product) and the combinatorial (lattice path) sides. Making this identification precise is the central task.

6. **Wheeler's puzzle programme: progressive generalisation.** Wheeler (with Zinn-Justin and others) has systematically extended the R-matrix → puzzle correspondence to increasingly general families of symmetric functions. The pattern is always the same: represent the symmetric functions as partition functions of an integrable vertex model, then use the Yang–Baxter equation to derive puzzle-type product rules.

   The RTT = TTR construction. Given an R-matrix $R_{ab}(z) \in \mathrm{End}(V \otimes V)$ satisfying the Yang–Baxter equation

   $$R_{12}(x/y)\, R_{13}(x/z)\, R_{23}(y/z) = R_{23}(y/z)\, R_{13}(x/z)\, R_{12}(x/y),$$

   one builds L-matrices (local vertex weights) and a monodromy matrix $T_a(x) = \prod_{i\geqslant 0} L_a(x|\mathfrak{f}_i)$. The intertwining relation $R_{ab}(y/x)\, T_a(x)\, T_b(y) = T_b(y)\, T_a(x)\, R_{ab}(y/x)$ forces the transfer matrices to commute, which yields symmetric function identities as a consequence.

   The four stages:

   (a) **Hall polynomials and inverse Kostka** [32]: a rank-1 bosonic model (Fock space $|m\rangle$, $m \in \mathbb{Z}_{\geqslant 0}$) for Hall polynomials; a fermion–boson model for generalised inverse Kostka polynomials. Both give "right-triangle puzzles" with polynomial t-weights. At $t = 0$, recovers the classical LR puzzle rule.

   (b) **LR for Grothendieck polynomials** [33]: embeds a 5-vertex model (for double Grothendieck polynomials) into a rank-2 eleven-vertex model. The Yang–Baxter equation yields puzzle rules for equivariant K-theory of Grassmannians, generalising Knutson–Tao–Vakil.

   (c) **Colored fermionic vertex models** [1]: vertex models from $U_q(\widehat{\mathfrak{sl}}(1|n))$ with fusion of the fundamental R-matrix. LLT polynomials arise as partition functions; new proofs of Cauchy identities, branching rules, and formulas for nonsymmetric Macdonald polynomials follow.

   (d) **Spin Hall–Littlewood functions** [6]: lattice model from $U_q(\widehat{\mathfrak{sl}}(r+1))$ gives puzzle formulae for the structure constants of Borodin's spin Hall–Littlewood functions—a one-parameter generalisation of Schur functions. The proof applies Yang–Baxter in the style of Cauchy-identity proofs (simpler than the "train argument" of earlier work).

Observation 12.1 (Boltzmann weights across the four stages)**.** The combinatorial structure—partitions interlacing through a grid, with two particle species (+/−) at each edge—is *identical* in all four stages. What changes is the **weight function** at each vertex, parametrised by the *cumulative charge* c (= running count of particles in the column above the tile):

Stage	**Cross** $+\leftrightarrow-$	**Cross** $-\leftrightarrow+$	**Parameters**
Schur	$\{0,1\}$	$\{0,1\}$	none
Hall–Littlewood	$1-t^c$	t^c	t
Inverse Kostka	$\delta_{c,1}(1-t)$	$(-t)^c$	t
Grothendieck	$1-z$	z	$z=y_i/y_j$ (equiv.)
Spin Hall–Littlewood	q-hyper. in (q,s)	q-hyper.	q,s (spin param.)

All other tile types (both particles pass straight through) have weight 1. In each case, the structure constant (Hall polynomial, Grothendieck LR coefficient, etc.) is the **sum of products** of these local weights over all legal tilings:

$$f^{\bar{\lambda}}_{\bar{\mu}\bar{\nu}}(t)=\sum_{P\text{ puzzle}}\prod_{\text{tile }\tau\in P}w_\tau(t),\qquad c^K_{\lambda\mu\nu}(y)=\sum_P\prod_\tau w_\tau(y_i/y_j).$$

At $t=0$ (resp. $z=0$, $s=0$), the cross-weights become $\{0,1\}$ and all formulas reduce to the classical LR puzzle count.

Comparison with our Pieri coefficients. The cumulative charge c in Wheeler's puzzles tracks "how many particles sit above this vertex"—equivalently, how deep a cell is relative to the partition boundary. This is the same data encoded by the *arm and leg lengths* $a(s),\ell(s)$ in the Macdonald Pieri coefficient

$$\varphi_{\lambda/\mu}(q,t)=\prod_{s\in C_\lambda(\mu)}\frac{1-q^{a_\lambda(s)}t^{\ell_\lambda(s)+1}}{1-q^{a_\lambda(s)+1}t^{\ell_\lambda(s)}},$$

where $C_\lambda(\mu)$ is the set of corner cells of the strip λ/μ (eq. (8), Section 5). The structural parallel is:

Wheeler's puzzles	**Our growth diagram**
Cumulative charge c at tile	Arm/leg $a(s),\ell(s)$ at cell s
Cross-weight $1-t^c$	Pieri factor $\frac{1-q^at^{\ell+1}}{1-q^{a+1}t^\ell}$
Product over tiles in puzzle	Product over corner cells of strip
Sum over all puzzles	(single term: the strip is deterministic)

The key difference is in the last row: in the unweighted (Schur) case, the Burge local rule is a *bijection*, so the "sum over puzzles" has a single term. In the (q,t)-weighted case, the local rule becomes a *weighted sum* over multiple outputs at each vertex, and the Pieri coefficient is the accumulated weight of the unique path through the growth diagram. Making this dictionary precise is Question 12.3.

Remark 12.2 (Connection to our programme). Wheeler's work traces a *monotone path* from Schur (unweighted puzzles, all Boltzmann weights $\in \{0, 1\}$) to Macdonald (fully (q, t)-weighted vertex models):

$$\text{Schur} \xrightarrow{t} \text{Hall–Littlewood} \xrightarrow{q} \text{LLT / Macdonald.}$$

Each stage uses the same puzzle/tiling combinatorics but assigns progressively richer Boltzmann weights. Since we already have a Macdonald analog of Borodin's identity [15] (Section 5), what remains is Question 12.3: expressing the Pieri coefficient $\varphi_{\lambda/\mu}(q, t)$ as a product of R-matrix entries over vertices of the crossing-path model. This would upgrade the algebraic identity to a vertex-model partition function and open the door to positivity via Wheeler's framework.

Consequence for cylindric partitions. The trace formula writes $\langle s_\kappa, \mathrm{GK}_c \rangle$ as a sum over cyclic chains of partitions, with one LR factor per step. If each LR factor acquires a (q, t)-weight via the R-matrix, then:

$$\langle s_\kappa, \mathrm{GK}_c^{\mathrm{Mac}} \rangle = \sum_{\text{cyclic chains}} \prod_{k=0}^{T-1} c^{(q,t)}_{\lambda^{(k)},\lambda^{(k+1)},\kappa_k},$$

where each $c^{(q,t)}$ is a vertex-model partition function. If these (q, t)-LR coefficients are *polynomials* with nonneg coefficients, then so is their product—giving Macdonald Schur positivity of GK_c as a consequence of a much cleaner *local* positivity statement.

Question 12.3. Make the identification between the Macdonald Pieri coefficient $\varphi_{\lambda/\mu}$ and the accumulated six-vertex R-matrix along a horizontal strip precise. Concretely: given a horizontal strip λ/μ viewed as a lattice path configuration, does the product of R-matrix Boltzmann weights $\prod_v R(z_v)$ over all vertices v in the strip equal $\varphi_{\lambda/\mu}(q, t)$ (after appropriate identification of spectral parameters with q, t)?

Remark 12.4 (State of play). The situation decomposes cleanly into three layers:

Done. **Algebraic identity.** The (q, t)-Borodin identity (7) (Section 5, proved in [15]) gives a Macdonald analog of Borodin's product formula. The Macdonald Pieri coefficients $\varphi_{\lambda/\mu}(q, t)$ and $\psi_{\lambda/\mu}(q, t)$ already serve as local weights at each vertex of the growth diagram—the proof telescopes identically to the Schur case.

Gap. **Combinatorial identification** (Question 12.3). Show that $\varphi_{\lambda/\mu}(q, t)$ factorises as a product of R-matrix Boltzmann weights $\prod_v R(z_v)$ over the vertices of the horizontal strip λ/μ in the crossing-path model. This would recast the (q, t)-Borodin identity as a lattice-model partition function and make the local structure explicit.

Available. **Wheeler's machinery.** Wheeler–Zinn-Justin [32, 33] and Aggarwal–Borodin–Wheeler [1] have built the R-matrices that produce (q, t)-weighted puzzles for Hall–Littlewood, Grothendieck, LLT, and Macdonald polynomials. Once the Gap is closed, their vertex-model technology applies directly: positivity of cylindric Macdonald–Schur coefficients would follow from *local* positivity of the Boltzmann weights at each vertex, rather than from global cancellation arguments.

Remark 12.5 (Literature). A RAG index of 33 papers (2582 chunks) citing Zinn-Justin [34] is at `data/puzzles/` with a knowledge graph (249 concepts, 196 edges) and interactive D3.js visualisation. Key papers: [34] (core puzzle paper), [9] (equivariant puzzles), [10, 11, 12] (Schubert puzzles I–III: R-matrices from quantum groups yield puzzle rules for Schubert structure

constants in cohomology, K-theory, and equivariant K-theory), [32] (Hall polynomials/puzzles), [31] (Hall–Littlewood t-boson), [33] (Grothendieck LR), [1] (colored fermionic models), [6] (spin Hall–Littlewood). Query: `python3 data/puzzles/rag.py "your question"`; knowledge graph: `python3 data/puzzles/kg_query.py "concept"`.

13 Rains–Warnaar (q,t)-Littlewood–Richardson conjectures

In the classical setting, for a complex semisimple Lie algebra $\mathfrak{g}$, the *Littlewood–Richardson coefficient* $c_{\mu\nu}^{\lambda}$ of type $\mathfrak{g}$ is the multiplicity of the irreducible module $V(\lambda)$ in the tensor product $V(\mu)\otimes V(\nu)$:

$$c_{\mu\nu}^{\lambda} = \dim \mathrm{Hom}_{\mathfrak{g}}\big(V(\lambda),\, V(\mu)\otimes V(\nu)\big).$$

Stembridge [25] classified all *multiplicity-free* tensor products, i.e. pairs (μ,ν) with $c_{\mu\nu}^{\lambda}\leqslant 1$ for all λ. For $\mathfrak{g}=\mathrm{B}_n$ with $(\mu,\nu)=(r\omega_n, s\omega_n)$, Okada [20] proved that $c_{\mu\nu}^{\lambda}=1$ for all dominant weights λ of the appropriate form, giving the identity

$$\mathrm{so}_{2n+1,\,(\frac{r}{2})^n}(x)\ \mathrm{so}_{2n+1,\,(\frac{s}{2})^n}(x) \;=\; \sum_{\substack{\lambda\\ \lambda_1\leqslant \min\{r,s\}}} \mathrm{so}_{2n+1,\,(\frac{r+s}{2})^n-\lambda}(x),$$

and an analogous formula for C_n with symplectic Schur functions.

Rains and Warnaar [22] conjectured (q,t)-lifts of both identities to the level of Macdonald–Koornwinder polynomials.

Open Problem 6 ([22], B_n type). *For $x=(x_1,\dots,x_n)$ and r,s nonnegative integers, prove that*

$$P^{(\mathrm{B}_n,\mathrm{B}_n)}_{(\frac{r}{2})^n}(x;\,q,t,t)\;P^{(\mathrm{B}_n,\mathrm{B}_n)}_{(\frac{s}{2})^n}(x;\,q,t,t) \;=\; \sum_{\lambda} b^{\mathrm{el}}_{\lambda;\,r,s}(q,t)\;P^{(\mathrm{B}_n,\mathrm{B}_n)}_{(\frac{r+s}{2})^n-\lambda}(x;\,q,t,t), \tag{18}$$

where the (q,t)-LR coefficient is

$$b^{\mathrm{el}}_{\lambda;\,m_1,m_2}(q,t) := b^{\mathrm{el}}_{\lambda}(q,t)\prod_{\substack{s\in\lambda\\ l'(s)\text{ even}}}\frac{1-q^{m_1-a'(s)}t^{l'(s)}}{1-q^{m_1-a'(s)-1}t^{l'(s)+1}}\cdot\frac{1-q^{m_2-a'(s)}t^{l'(s)}}{1-q^{m_2-a'(s)-1}t^{l'(s)+1}}\prod_{\substack{s\in\lambda\\ l(s)\text{ even}}}\frac{1-q^{m_1+m_2-\hat{a}(s)-1}t^{\hat{l}(s)+1}}{1-q^{m_1+m_2-\hat{a}(s)}t^{\hat{l}(s)}}.$$

Here $b^{\mathrm{el}}_{\lambda;\,m_1,m_2}(q,t)=0$ unless $\lambda_1\leqslant\min\{m_1,m_2\}$, and the large-$s$ limit recovers the bounded Littlewood identity of [22, Theorem 1.1].

Open Problem 7 ([22], C_n type). *For $x=(x_1,\dots,x_n)$ and r,s nonnegative integers, prove that*

$$P^{(\mathrm{C}_n,\mathrm{B}_n)}_{r^n}(x;\,q,t,qt)\;P^{(\mathrm{C}_n,\mathrm{B}_n)}_{s^n}(x;\,q,t,qt) \;=\; \sum_{\lambda\text{ even}} b^{\mathrm{oa}}_{\lambda;\,r,s}(q,t)\;P^{(\mathrm{C}_n,\mathrm{B}_n)}_{(r+s)^n-\lambda}(x;\,q,t,t), \tag{19}$$

where $b^{\mathrm{oa}}_{\lambda;\,m_1,m_2}(q,t)=0$ unless $\lambda_1\leqslant\min\{2m_1,2m_2\}$, and the sum is over even partitions.

Remark 13.1 (Connection to our work). The large-s limits of (18) and (19) recover the bounded Littlewood identities of Theorems 4.1 and 1.1 of [22]. These in turn specialise at $q=0$ to bounded identities for Hall–Littlewood polynomials, whose character specialisations yield the Andrews–Gordon identities underlying Conjecture 2.7. The (q,t)-LR coefficients $b^{\mathrm{el}}_{\lambda;\,r,s}$ and $b^{\mathrm{oa}}_{\lambda;\,r,s}$ are explicit closed-form products over squares of λ—exactly the type of formula one would hope a Macdonald statistic (Section 12) on puzzles or hives could produce. If Open Problems 6–7 could be proved via an integrable lattice model, the Boltzmann weights would give a natural (q,t)-statistic on the combinatorial objects computing LR coefficients.

14 Cyclic descent injection ($n = 1$ partial result)

Remark 14.1 (The $n = 1$ case is not trivial). One might expect $Q_{1,c}(q)$-positivity to be easy, but it is not. The formula gives $Q_1 = (1-q)\, g_1(q) - q$, an alternating expression. The coefficient of q^w is the number of max-1 CPPs of weight w minus the number of weight $(w-1)$, minus a correction. This requires showing that the triple count is weakly increasing (for w large enough)—precisely what the cyclic descent injection below achieves.

For max-1 CPPs (single triples (l_1, l_2, l_3) with $l_1 + l_2 + l_3 = w$ and slack $s_i = l_{i-1} + c_i - l_i$), define the *cyclic descent set* $\mathrm{CycDes}(s) = \{k : s_k > s_{k-2 \bmod 3}\}$. The injection rule: increment component k where $k = \arg\min\{s_{k-2 \bmod 3} : k \in \mathrm{CycDes}(s)\}$.

Observation 14.2 (Verified: 126 profiles, $d \le 8$, $d \not\equiv 0 \pmod 3$)**.** This gives a valid injection from weight-$(w-1)$ to weight-w triples. The fixed points at weight $w \ge 2$ match the coefficients of $h_1(q) = Q_{1,c}(q)$.

For $d \equiv 0 \pmod 3$: the balanced slack $(d/3, d/3, d/3)$ has empty cyclic descent set, making the rule inapplicable. For $d = 3$ the triple count oscillates $(3, 3, 4, 3, 3, 4, \ldots)$, precluding any uniform injection. (`scripts/test_cyclic_rule.py`.)

15 Dead ends

1. **Soft positivity** (disproved): $f(z, q) = 1 + zq$ has nonneg g_m but $Q_n < 0$. The cylindric structure is essential.
2. **Kostka–Foulkes decomposition**: $Q_{n,c}(q)$ does not decompose as a nonneg sum of $K_{\lambda,\mu}(q)$. (`scripts/analyze_Q.py`.)
3. **Multivariate** Q_n ($q^k \to q_k$): positivity fails for $n \ge 2$ with negative integer coefficients.
4. **E–H in** QSym: both $\prod(1 - tz_k) \cdot \widetilde{\mathrm{GK}}_c$ and $\prod(1 - z_k) \cdot \widetilde{\mathrm{GK}}_c$ have negatives.
5. **Combinatorial involution**: profile-preserving operators move a variable number of boxes per CPP, but involutions require weight preservation. Dead end.
6. **Korff–Palazzo h-positivity** [14]: their "h" (cylindric complete symmetric function in the $\{h_\nu\}$ basis) differs from our $h_m = (q)_m \cdot g_m$. Their positivity is in the wrong direction: $Q_n \ge 0 \Rightarrow h_m \ge 0$ trivially by (15), but $h_m \ge 0$ does not imply $Q_n \ge 0$.
7. **Dominance relations**: $D_m = h_m - q \cdot h_{m-1} \ge 0$ and $R_n = h_n - Q_n \ge 0$ are both *consequences* of $Q_n \ge 0$, not independent.
8. **Path-index involution on** (S, ν'): In the two-level formulation (16), removing k from S decreases $\sigma(S)$ by k, so weight-preservation requires adding exactly k boxes to ν'. The natural idea: interpret $k \in S$ as selecting the k-th lattice path in a CPP with max n, and define the involution by an operation on that path (adding it back, splitting a level, etc.). **This fails** because:
   - Adding/removing path k changes the weight by a configuration-dependent amount (the area of the strip between adjacent paths), not by k.
   - A local rule (Burge / growth diagram) **changes the profile**—it maps between $\mathrm{CPP}(c)$ and $\mathrm{CPP}(c')$ for $c' \ne c$.

- Relabelling (inserting a value level at position k) changes the weight by the number of cells with value $\geq k$, which varies per ν'.

The core obstruction: k is a property of S (a path index), but the weight change from modifying ν' depends on the geometry of ν'. These two quantities are decoupled.

16 Most promising future directions

1. **Integrable vertex model / Macdonald statistic** (Section 12): the most promising direction for connecting the (q,t)-generalisation to the positivity conjecture. Unifies the growth-diagram proof of Borodin's identity with Zinn-Justin's integrable tiling framework.

2. **0-Hecke module construction**: L-positivity $\Leftrightarrow$ quasi-symmetric characteristic of an $H_n(0)$-module. Explicitly constructing this module for CPPs would give a representation-theoretic proof of L-positivity and might connect to $Q_{n,c}$ via $H_n(0)$-representation theory.

3. **P-partition poset for CPPs**: If the interlacing conditions can be modelled as a labeled poset (handling the cyclic closure by unrolling), L-positivity follows from Stanley–Gessel theory, and the L-coefficients count linear extensions by descent set.

4. **Extending cyclic descent to $n \geq 2$**: The injection works for $n = 1$. A multi-path analog (injection on n-tuples of non-intersecting paths) could prove $Q_{n,c} \geq 0$ directly.

5. **Affine crystal structure**: Tingley [28] defines Kashiwara operators $\tilde{e}_i, \tilde{f}_i$ on CPPs preserving profile. These change weight by one box (a root α_i) but do not preserve max entry. Understanding the interplay between crystal operators and the max-entry filtration may reveal the right framework.

6. **Demazure atoms and key polynomials**: Lifting $\widetilde{\mathrm{GK}}_c$ from QSym to the polynomial ring as a sum of Demazure atoms (Mason [19]) could encode the max-entry information lost in QSym.

Computational tools

All computations use SageMath 10.7. Key scripts in `scripts/` (importing from the `cpp_lib/` package):

`compute_gk.py`	GK enumeration, Borodin validation
`verify_conjecture.py`	$Q_{n,c}(q)$ positivity check
`analyze_Q.py`	Kostka–Foulkes comparison, unimodality
`investigate_involution.py`	E–H factorisation, q-binomial
`test_cyclic_rule.py`	Cyclic descent injection
`analyze_qsym_cancellation.py`	Cross-cancellation anatomy
`verify_partial_qsym.py`	Partial QSym at fixed max
`macdonald_schur.py`	Macdonald $P/Q/\tilde{H}$ Schur positivity

TODO

Writing tasks

1. **TikZ diagrams in this document.** DONE — Both CPP visualisations (labelled cylindric diagram and stacked cubes 3D) imported into Remark 1.2.

2. **Explicit local rule formula.** DONE — Burge rule (both $\mathfrak{D}$ and $\mathfrak{U}$) with worked example added to Section 1.5.

3. **Growth diagram figure.** DONE — Example 1.11 added to Section 1.6.

4. **Cylindric Schur function.** DONE — Definitions 1.5 and 1.6 now give the full formal definition (cylindric loop, skew diagram, periodicity condition).

5. **Charge algorithm: worked example.** DONE — Example 3.7 added after Definition 3.6.

6. A_r **generalisation.** DONE — Conjecture 8.4 states Warnaar's Conjecture 8.4 for general rank r, with the $q = 1$ evaluation formula (12).

Computations to run

1. $d \equiv 0 \pmod 3$**: does $Q_{n,c}$ go negative?** DONE — Computed $Q_{n,c}(q)$ for $n = 0, \ldots, 3$ with max_w = 20 for six profiles: $d = 3$ ($c = (1,1,1), (2,1,0), (3,0,0)$) and $d = 6$ ($c = (2,2,2), (3,2,1), (4,1,1)$). **All coefficients are nonneg.** However:

   - $Q_{n,c}(1)$ is *not* a perfect power of $Q_{1,c}(1)$ for $n \geq 3$ (e.g. $c = (1,1,1)$: $Q_1(1) = 7$, $Q_2(1) = 49 = 7^2$, $Q_3(1) = 277 \neq 7^3$).
   - Different profiles with the same d give *different* $Q_3(1)$ (e.g. $d = 3$: $(1,1,1) \to 277$, $(2,1,0) \to 218$, $(3,0,0) \to 134$).

   So the conjecture's exclusion of $d \equiv 0 \pmod 3$ is not about positivity failing, but about the evaluation formula $Q_n(1) = K^n$ breaking. See `scripts/test_d0mod3.py`.

2. **Macdonald Schur positivity.** DONE — `scripts/macdonald_schur.py` confirms: $P/Q \to s$ gives *rational functions* (ill-posed); $\tilde{H}_\mu \to s$ gives $\mathbb{N}[q,t]$ coefficients (Haiman's theorem, verified $|\mu| \leq 4$); skew Q products (appearing in the Borodin identity) are rational.

3. (q,t)**-analog of $Q_{n,c}$.** DONE — The (q,t)-Borodin identity (Section 5) gives rational Schur coefficients in the P/Q basis, making a direct (q,t)-Q polynomial ill-posed. See Section 5.3.

4. $\tilde{H}$**-positivity.** Check whether GK_c expands with nonneg coefficients in the modified Macdonald basis $\{\tilde{H}_\mu\}$.

Mathematical questions

1. **Prove L-positivity of $\widetilde{\mathrm{GK}}_c$.** Currently only verified computationally (Observation 2.4). The P-partition argument works degree by degree; extend to the full generating function.

2. **Find a descent statistic on CPPs.** Conjecture 3.11 predicts a statistic $\alpha(\mathfrak{c})$ such that $\widetilde{\mathrm{GK}}_c = \sum L_{\alpha(\mathfrak{c})}$. Candidates: reading word of the labelled cylindric diagram?

3. **Explore toric P-partitions for L-positivity.** Adin–Gessel–Reiner–Roichman (arXiv:1811.05440) define *toric P-partition enumerators* for toric DAGs, which expand positively in *cyclic* quasi-symmetric functions F_α^{cyc}. CPPs have intrinsic cyclic structure (the profile wraps around $\mathbb{Z}/3\mathbb{Z}$), but the cells of a CPP form a variable-size set rather than a fixed finite poset—so the framework does not apply directly. Three approaches to investigate:

   (a) *Row-by-row toric factorisation.* View each row of a CPP as a 1-dimensional "toric" path (the interlacing condition is cyclic in the profile index). Each row yields a toric order on a bounded alphabet; the multi-row condition becomes non-intersecting toric P-partitions.

   (b) *Enriched toric D-partitions (Chow 2001, Petersen 2005).* The enriched P-partition theory refines descent to peak/valley data. An enriched toric analogue might accommodate the extra symmetry in the CPP interlacing.

   (c) 0-*Hecke module directly.* Skip posets entirely; construct an explicit $H_n(0)$-module whose quasi-symmetric characteristic is $\widetilde{\mathrm{GK}}_c$. Tewari–van Willigenburg [27] give the abstract equivalence L-positive $\Leftrightarrow$ $H_n(0)$-module, but no explicit module for CPPs is known.

4. **Quasisymmetric Littlewood–Richardson rule.** Haglund–Luoto–Mason–van Willigenburg [8] define *quasisymmetric Schur functions* that refine classical Schur functions and prove a quasisymmetric Pieri rule. However, a combinatorial rule for the product of two generic quasisymmetric Schur functions (a "quasisymmetric LR rule") remains open [8, p. 23]. Since $\widetilde{\mathrm{GK}}_c$ lives in QSym, such a rule could give finer structural information about the Schur coefficients of CPP generating functions.

5. **Combinatorial description of Schur coefficients.** Done — The crossing-path model (Section 7) reformulates $\langle s_\kappa, \mathrm{GK}_c \rangle$ as a single count of crossing-path configurations on the cylinder (Proposition 7.2). Next: define statistics on these configurations.

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